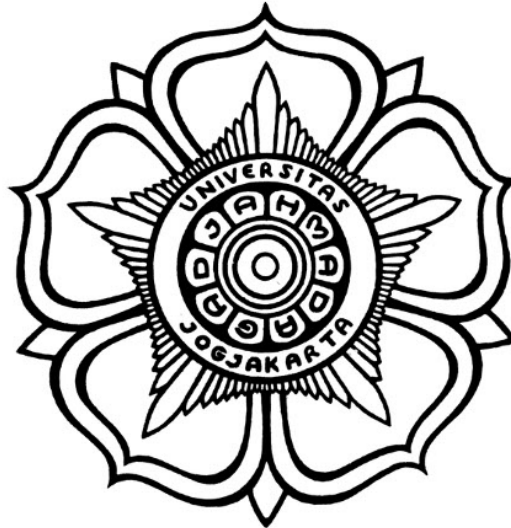


**RAYCASTED: A RAYCAST END-TO-END DETECTION MODEL
FOR LIGHTWEIGHT AND BOUNDARY-AWARE NUCLEI
DETECTION IN HISTOPATHOLOGY IMAGES**

SKRIPSI



**THE SUSTAINABLE DEVELOPMENT GOALS
Industry, Innovation and Infrastructure
Affordable and Clean Energy
Climate Action**

Written by:

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BIOMEDICAL ENGINEERING PROGRAM

**DEPARTMENT OF ELECTRICAL ENGINEERING AND
INFORMATION TECHNOLOGY
FACULTY OF ENGINEERING UNIVERSITAS GADJAH MADA
YOGYAKARTA
2026**

ENDORSEMENT PAGE

RAYCASTED: A RAYCAST END-TO-END DETECTION MODEL FOR LIGHTWEIGHT AND BOUNDARY-AWARE NUCLEI DETECTION IN HISTOPATHOLOGY IMAGES

THESIS

Proposed as A Requirement to Obtain
Undergraduate Degree (*Sarjana Teknik*)
in Department of Electrical Engineering and Information Technology
Faculty of Engineering
Universitas Gadjah Mada

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PREFACE

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Akhir kata penulis berharap semoga skripsi ini dapat memberikan manfaat bagi kita semua, aamiin.

Catatan: setiap nama yang dituliskan boleh disertai dengan alasan berterima kasih.

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NOMENCLATURE AND ABBREVIATION

[SAMPLE]

b	=	bias
$K(x_i, x_j)$	=	fungsi kernel
y	=	kelas keluaran
C	=	parameter untuk mengendalikan besarnya pertukaran antara penalti variabel slack dengan ukuran margin
L_D	=	persamaan Lagrange dual
L_P	=	persamaan Lagrange primal
$\mathbf{w}$	=	vektor bobot
$\mathbf{x}$	=	vektor masukan
ANFIS	=	Adaptive Network Fuzzy Inference System
ANSI	=	American National Standards Institute
DAG	=	Directed Acyclic Graph
DDAG	=	Decision Directed Acyclic Graph
HIS	=	Hue Saturation Intensity
QP	=	Quadratic Programming
RBF	=	Radial Basis Function
RGB	=	Red Green Blue
SV	=	Support Vector
SVM	=	Support Vector Machines

INTISARI

Intisari ditulis menggunakan bahasa Indonesia dengan jarak antar baris 1 spasi dan maksimal 1 halaman. Intisari sekurang-kurangnya berisi tentang latar belakang dan tujuan penelitian, metodologi yang digunakan, hasil penelitian, kesimpulan dan implikasi, dan Kata kunci yang berhubungan dengan penelitian.

Kata Kunci ditulis maksimal 5 kata yang paling berhubungan dengan isi skripsi. Silakan mengacu pada ACM / IEEE *Computing classification* jika Anda adalah mahasiswa Sarjana TI <http://www.acm.org/about/class/> atau mengacu kepada IEEE keywords http://www.ieee.org/documents/taxonomy_v101.pdf jika Anda berasal dari Prodi Sarjana TE.

Kata kunci : Kata kunci 1, Kata kunci 2, Kata kunci 3, Kata kunci 4, Kata kunci 5

Contoh Abstrak Teknik Elektro:

"Penelitian ini bertujuan untuk mengembangkan sistem pengendalian suhu ruangan dengan menggunakan microcontroller. Metodologi yang digunakan adalah desain sirkuit, implementasi sistem pengendalian, dan pengujian performa. Hasil penelitian menunjukkan bahwa sistem pengendalian suhu ruangan yang dikembangkan mampu mengendalikan suhu ruangan dengan akurasi sebesar $\pm 0,5^{\circ}\text{C}$. Kesimpulan dari penelitian ini adalah sistem pengendalian suhu ruangan yang dikembangkan efektif dan efisien.

Kata kunci: microcontroller, sistem pengendalian suhu, akurasi."

Contoh Abstrak Teknik Biomedis:

"Penelitian ini bertujuan untuk mengevaluasi keefektifan prototipe alat pemantau denyut jantung berbasis elektrokardiogram (ECG) untuk pasien jantung. Metodologi yang digunakan meliputi desain dan pembuatan prototipe, pengujian dengan pasien, dan analisis data. Hasil penelitian menunjukkan bahwa prototipe alat pemantau denyut jantung berbasis ECG memiliki akurasi yang baik dan mampu memantau denyut jantung pasien secara efektif. Kesimpulan dari penelitian ini adalah prototipe alat pemantau denyut jantung berbasis ECG merupakan solusi yang efektif dan efisien untuk memantau pasien jantung.

Kata kunci: elektrokardiogram, alat pemantau denyut jantung, akurasi."

Contoh Abstrak Teknologi Informasi:

"Penelitian ini bertujuan untuk mengevaluasi keamanan dan privasi pengguna aplikasi media sosial terpopuler. Metodologi yang digunakan meliputi analisis kebijakan privasi dan pengaturan keamanan, pengujian penetrasi, dan survei pengguna. Hasil penelitian menunjukkan bahwa beberapa aplikasi media sosial memiliki kebijakan privasi yang kurang jelas dan rendahnya tingkat keamanan. Kesimpulan dari penelitian ini adalah pentingnya meningkatkan kebijakan privasi dan tingkat keamanan pada aplikasi media sosial untuk melindungi privasi dan data pengguna.

Kata kunci: media sosial, keamanan, privasi, pengguna."

ABSTRACT

Abstract ditulis italic (miring) menggunakan bahasa Inggris dengan jarak antar baris 1 spasi dan maksimal 1 halaman. Abstract adalah versi Bahasa Inggris dari intisari. Abstract dapat ditulis dalam beberapa paragraf. Baris pertama paragraph harus menjorok ke dalam sekitar 1 cm. Tidak disarankan menggunakan mesin penerjemah melainkan tulis ulang.

Keywords : Keyword 1, Keyword 2, Keyword 3, Keyword 4, Keyword 5

CHAPTER I

INTRODUCTION

1.1 Background

Vision-based deep learning models have become a standard tool for analyzing the tumor microenvironment (TME) in whole-slide images (WSIs) of histopathology tissue. Among the many biomarkers extracted from the TME, tumor-infiltrating lymphocytes (TILs) density is an established prognostic indicator in multiple cancer types [1]. The International TILs Working Group recommends scoring TILs by estimating the percentage of stromal area occupied by lymphocytes, which is fundamentally an *area-based* measurement that depends on accurately delineating cell boundaries rather than merely locating cell centroids. Historically, segmentation-based approaches have been the standard for this task, as they assign a class label to every pixel and thereby produce the per-cell boundary maps needed for area measurement.

However, segmentation methods face two fundamental limitations when applied to nuclei in histopathology images. **Semantic segmentation** assigns a single class label to every pixel, which is sufficient for macro-level area measurements such as overall tissue region delineation or total TIL density estimation. Yet it fails entirely when accurate individual cell counts are required: because touching or overlapping nuclei share contiguous pixel regions, the model merges them into a single connected component, making it impossible to distinguish where one cell ends and another begins. **Instance segmentation** attempts to address this by separating individual instances, but introduces its own challenges. The *overlap problem* arises because in a two-dimensional projection, a single pixel cannot simultaneously belong to two overlapping objects, forcing mutually exclusive pixel assignments that distort cell morphology at contact boundaries. Additionally, the *computational bottleneck* is significant: the model must classify every foreground and background pixel across potentially massive whole-slide images, making instance segmentation methods such as U-Net [2] and Mask R-CNN [3] prohibitively expensive at clinical scale.

Object detection models offer a lighter alternative. By outputting a fixed set of parameters per object rather than a dense pixel-wise prediction, detection frameworks such as Faster R-CNN [4] and YOLO [5] naturally separate instances, incur lower computational overhead, and excel at counting tasks. However, detection methods suffer from two critical shortcomings. The *boundary problem* is that bounding boxes disregard the geometric boundaries of cells, overestimating cell size and destroying the morphological data necessary for area-based TILs scoring. The *density problem* is that detection models rely on non-maximum suppression (NMS) as a post-processing step to eliminate duplicate

predictions, which forces a trade-off where valid overlapping cells in densely clustered regions are frequently suppressed. This is precisely where accurate counting matters most for clinical assessment.

Some recent methods have attempted to bridge these gaps. StarDist [6] represents cells as star-convex polygons, preserving boundary shape while maintaining per-cell separation, and HoVer-Net [7] uses horizontal and vertical distance maps to separate clustered nuclei. However, both rely on a two-stage pipeline of dense prediction followed by post-processing, and still use NMS, which risks undercounting in the very regions where accuracy matters most. LSP-DETR [8] addresses the NMS issue by adopting an end-to-end DETR-style framework with learned assignment, eliminating NMS entirely while representing nuclei as star-convex polygons. However, its transformer-based architecture imposes quadratic computational complexity that makes it impractical for resource-constrained edge deployment. CellPose-SAM [9] has demonstrated strong generalization for cellular segmentation, though its computational profile is not optimized for real-time inference on edge devices.

No existing method combines polygon-based cell representation with a single-stage, end-to-end fully convolutional network (FCN) detection architecture that eliminates NMS dependency while maintaining the lightweight computational profile necessary for edge deployment. This thesis addresses this gap by proposing **RayCastED**, a detection model heavily inspired by LSP-DETR [8] in its use of star-convex polygon predictions within an end-to-end, NMS-free framework, but designed from the ground up with a fully convolutional architecture. RayCastED reframes the star-convex representation as a *raycast* representation, defined by a centroid plus fixed-angle radial rays, integrated directly into a YOLO-based detection head with YOLO’s native assignment mechanism. By adopting a purely FCN-based architecture rather than a transformer, RayCastED maintains the lightweight profile needed for edge deployment through TensorRT and ONNX Runtime while remaining end-to-end and NMS-free, thereby combining the morphological accuracy of segmentation with the lightweight efficiency of detection.

1.2 Problem Statement

Based on the background discussed above, the following problems are identified:

1. **Semantic segmentation merges overlapping cells:** Pixel-wise classification merges touching or overlapping nuclei into a single connected component, making individual cell counts impossible and undermining the per-cell boundary delineation required for area-based TILs scoring.
2. **Instance segmentation is computationally prohibitive:** Classifying every foreground and background pixel across massive whole-slide images imposes a compu-

tational bottleneck that makes instance segmentation impractical at clinical scale, particularly in resource-constrained environments that require edge deployment.

3. **Detection discards boundary geometry:** Bounding box representations disregard cell morphological boundaries, overestimating cell size and destroying the geometric data necessary for area-based scoring, despite detection models offering the lightweight inference profile needed for practical deployment.
4. **NMS suppresses valid cells in dense regions:** Non-maximum suppression, used by both detection and polygon-based segmentation methods, systematically deletes valid overlapping predictions in densely clustered regions, precisely where accurate counting matters most for clinical assessment.
5. **Transformer-based NMS-free methods are too heavy for edge deployment:** End-to-end frameworks such as LSP-DETR eliminate NMS through learned assignment but rely on transformer architectures with quadratic computational complexity, making them impractical for resource-constrained edge devices.

1.3 Objectives

To address the identified problems, this thesis has the following objectives:

1. Develop **RayCastED**, a nuclei detection model that:
 - preserves cell boundary geometry through a raycast polygon representation (centroid plus fixed-angle radial rays), addressing the boundary problem of detection models (Problem 3);
 - eliminates dense pixel-wise classification by predicting per-cell polygons directly from a detection head, addressing both the merging issue of semantic segmentation (Problem 1) and the computational bottleneck of instance segmentation (Problem 2);
 - replaces NMS with a learned end-to-end assignment mechanism, preventing undercounting in densely clustered regions (Problem 4); and
 - operates on an FCN backbone to maintain a lightweight computational profile suitable for edge deployment via TensorRT and ONNX Runtime, avoiding the quadratic complexity of transformer-based methods (Problem 5).
2. Evaluate the performance of RayCastED by comparing it against prior nuclei detection and segmentation models on standard benchmarks.
3. Evaluate the performance of RayCastED on edge devices.

1.4 Research Scope

This research is bounded by the following scope limitations:

1. **Data characteristics:** The study uses only H&E stained histopathology images with approximately 0.25 MPP (microns per pixel) resolution from publicly available secondary datasets. The datasets encompass multiple tissue types/organs.
2. **Computational constraints:** Model evaluation is performed on RTX 5070ti hardware. Baseline comparisons are limited to models compatible with this hardware configuration.
3. **Validation approach:** The research employs computational validation using standard metrics. Pathologist or expert validation is not included in this study.
4. **Technical boundaries:** There are no limitations on nuclei types, sizes, or detection scope. No strict constraints are imposed on input patch sizes or memory usage during model training and inference.

1.5 Research Novelty

The novelty of this research lies in the development of RayCastED, which represents the first architecture that combines three key elements: (1) a raycast polygon representation for preserving cell boundary geometry without dense pixel-wise classification, (2) an FCN backbone for maintaining the lightweight computational profile required for edge deployment, and (3) an end-to-end learned assignment mechanism that eliminates NMS dependency. This unique combination bridges the gap between the morphological accuracy of segmentation and the lightweight efficiency of detection for nuclei analysis in histopathology images.

1.6 Research Contributions

The contributions of this research are as follows:

1. **RayCastED model implementation:** A complete implementation of the proposed polygon-based detection model with raycast representation, integrated into a YOLO-based detection head with end-to-end assignment mechanism.
2. **Performance benchmarks on standard datasets:** Comprehensive evaluation of RayCastED on publicly available histopathology datasets, comparing performance against baseline models including semantic segmentation, instance segmentation, object detection, and transformer-based approaches.
3. **Edge deployment validation:** Validation of RayCastED’s deployment capabilities on edge devices through TensorRT and ONNX Runtime conversion, demonstrating practical feasibility for resource-constrained clinical environments.

4. **Comparison against baseline models:** Systematic comparison with prior nuclei detection and segmentation methods to quantify improvements in boundary accuracy, cell counting in dense regions, and computational efficiency.
5. **Reproducible codebase:** A publicly available, well-documented implementation that enables reproducibility and provides a foundation for future research in polygon-based detection for computational pathology.

1.7 Presentation Structure

Sistematika penulisan berisi pembahasan apa yang akan ditulis di setiap bab. Sistematika pada umumnya berupa paragraf yang setiap paragraf mencerminkan bahasan setiap Bab. Bagian ini akan ditulis setelah seluruh bab dalam skripsi ini selesai disusun.

CHAPTER II

TINJAUAN PUSTAKA DAN DASAR TEORI

2.1 Literature Review

This section reviews prior work on nuclei detection and segmentation in histopathology images, focusing on the methods most relevant to the proposed RayCastED model.

2.1.1 StarDist

StarDist [6] introduced the star-convex polygon representation for cell detection, originally developed for fluorescence microscopy and later adapted to histopathology images [10]. StarDist predicts a set of radial distances from each cell centroid to its boundary at fixed angular intervals, producing a star-convex polygon that preserves boundary geometry. However, StarDist relies on a two-stage pipeline: it first generates a dense probability map and then applies non-maximum suppression (NMS) to filter overlapping predictions, which risks undercounting in densely clumped regions and introduces computational overhead from post-processing.

2.1.2 CPP-Net

CPP-Net [11] proposed a context-aware polygon proposal network for nuclei instance segmentation. Rather than regressing radial distances from a fixed grid, CPP-Net generates polygon proposals by learning to predict polygon vertices directly, using a context-aware module that aggregates multi-scale features to refine proposal quality. While CPP-Net demonstrates that polygon-based representations can achieve high segmentation accuracy, it operates within a dense prediction framework and relies on NMS-based post-processing to remove duplicate proposals, inheriting the same NMS dependency as StarDist.

2.1.3 HoVer-NeXt

HoVer-NeXt [12] extended the HoVer-Net [7] framework for nuclei segmentation and classification, optimizing it for computational efficiency in large-scale histopathology applications. HoVer-Net predicts horizontal and vertical distance maps from each pixel to the centroids of neighboring cells, which are then used to separate touching or overlapping instances through post-processing. HoVer-NeXt improves upon this pipeline with architectural refinements that reduce inference time while maintaining segmentation quality. However, the method remains fundamentally a dense segmentation approach that classifies every pixel and requires computationally expensive post-processing (watershed algorithms and morphological operations) to extract individual instance masks.

2.1.4 CellPose

CellPose [13] introduced a generalist framework for cellular segmentation that predicts a gradient flow field pointing from each pixel toward the nearest cell centroid, combined with a cell probability map. Trained on a diverse dataset of over 70,000 segmented objects across multiple image types, CellPose demonstrates strong cross-domain generalization without requiring model retraining. However, CellPose operates as a dense segmentation model that classifies every pixel, making it computationally expensive for whole-slide analysis. A subsequent extension, CellPose-SAM [14], combined CellPose with the Segment Anything Model (SAM) to achieve superhuman generalization, though the computational profile remains unsuitable for resource-constrained deployment.

2.1.5 LSP-DETR

LSP-DETR [8] represents the most direct precursor to this work. It introduced a fully end-to-end framework for nuclei segmentation that represents nuclei as star-convex polygons within a DETR-style transformer architecture. By using a learned bipartite assignment mechanism instead of NMS, LSP-DETR eliminates the undercounting problem in clumped regions. The authors also proposed a novel radial distance loss that allows overlapping nuclei segmentation to emerge naturally. LSP-DETR demonstrated strong generalization across tissues and was over five times faster than the next-fastest leading method. However, its transformer-based backbone imposes quadratic complexity, making it computationally demanding for edge deployment. RayCastED is heavily inspired by LSP-DETR’s use of star-convex polygon predictions within an end-to-end, NMS-free framework, but is designed from the ground up with a fully convolutional architecture to maintain a lightweight computational profile suitable for resource-constrained environments.

Table 2.1 summarizes the key characteristics of the reviewed methods, highlighting how each method addresses (or fails to address) the core challenges identified in the background.

Table 2.1. Comparison of prior nuclei detection and segmentation methods. The **Fully Convolutional** column indicates whether the method uses a pure FCN backbone (no transformer or attention-based components).

Method	Approach	End-to-End	Fully Convolutional	Post-processing
StarDist [6, 10]	Detection	×	✓	NMS
CPP-Net [11]	Instance Seg.	×	✓	NMS
HoVer-NeXt [7, 12]	Instance Seg.	×	✓	Watershed + Morphological
CellPose [13, 14]	Instance Seg.	×	✓	NMS
LSP-DETR [8]	Detection	✓	×	None

As Table 2.1 shows, no existing method simultaneously achieves all three prop-

erties: end-to-end detection, polygon-based boundary representation, and a fully convolutional architecture. StarDist, CPP-Net, HoVer-NeXt, and CellPose all rely on FCN backbones but require handcrafted post-processing. LSP-DETR is end-to-end through bipartite matching but relies on a transformer backbone. RayCastED is designed to fill this gap by combining the end-to-end bipartite assignment of LSP-DETR with a purely FCN-based architecture.

2.2 Theoretical Basis

2.2.1 Histopathological Foundations

Histopathology is the study of tissue disease through microscopic examination of tissue samples. In clinical practice, tissue specimens are collected through biopsy or surgical resection and processed for microscopic analysis. The tissue undergoes fixation (typically in formalin) to preserve cellular structures, followed by embedding in paraffin wax, sectioning into thin slices, and mounting on glass slides for staining and visualization.

The cell nucleus is a membrane-bound organelle containing genetic material and serves as the primary structure of interest in histopathological analysis. Nuclear morphology, including size, shape, texture, and spatial distribution, provides critical diagnostic information. In cancer diagnosis, malignant cells often exhibit nuclear atypia: enlarged nuclei, irregular shapes, hyperchromasia (increased staining intensity), and abnormal mitotic figures.

2.2.1.1 Nuclear Morphology and Shape Characteristics

The shape of cell nuclei varies considerably across tissue types and pathological states, but the majority of nuclei encountered in routine H&E-stained histopathology sections exhibit a defining geometric property: they are **approximately convex**. Healthy epithelial nuclei are typically round or slightly oval, with smooth contours and minimal indentations. Lymphocytes are among the most regular, appearing as compact circles with diameters of approximately 5–8 μm . Fibroblast and endothelial nuclei tend to be elongated (spindle-shaped), but remain convex with no concavities. Even in malignant tissue, where nuclear pleomorphism introduces greater variability in size and eccentricity, most nuclei retain a fundamentally convex silhouette. Concavities do arise in practice, most commonly at the boundary between two overlapping nuclei where the cells appear to “press into” one another, but these are artifacts of the 2D projection rather than intrinsic features of the nuclear membrane itself.

This near-ubiquitous convexity has an important implication for automated representation. Because concave shapes are rarely encountered in their undistorted form, nuclear boundaries can be fully described by a centroid and a set of radial distances measured from that centroid to the boundary at regular angular intervals. This radial

representation captures the complete boundary geometry with a compact set of parameters, avoiding both the redundancy of dense per-pixel segmentation and the information loss of axis-aligned bounding boxes. The few cases where concavities do appear, primarily at overlap boundaries, are artifacts of the 2D projection and can be addressed by representing each overlapping nucleus independently rather than forcing a single non-overlapping mask.

2.2.1.2 The Overlapping Nuclei Problem

A fundamental challenge in nuclei analysis is that nuclei in tissue sections are densely packed and frequently overlap, forming clumped clusters where individual boundaries are difficult to distinguish even for expert pathologists. This overlapping occurs because tissue sections are typically cut at a thickness of 3–5 μm , which is comparable to or smaller than the diameter of many nuclei (typically 5–10 μm). As a result, multiple nuclei at different depths within the section are projected onto the same 2D image plane, creating the appearance of clumped or touching nuclei. This is not merely an artifact of imaging resolution but an inherent property of projecting 3D tissue architecture onto a 2D plane, motivating the development of automated methods capable of both accurately delineating individual nuclear boundaries and correctly separating touching or overlapping nuclei.

2.2.1.3 Area-based Measurements

Tumor-infiltrating lymphocytes (TILs) are immune cells, primarily lymphocytes and plasma cells, that have migrated into the tumor microenvironment. The presence and density of TILs reflect the host immune response against the tumor and have been established as a prognostic biomarker in multiple cancer types. Higher TILs levels are generally associated with improved outcomes, particularly in triple-negative and HER2-positive breast cancers, where they predict better response to chemotherapy and longer survival.

The International TILs Working Group, convened by Salgado et al. [1], established standardized guidelines for TILs assessment in clinical practice. The recommended metric is **stromal TILs (sTILs)**, defined as the proportion of stromal area occupied by mononuclear inflammatory cells within the invasive tumor border:

$$\text{sTILs (\%)} = \frac{\text{Area of mononuclear inflammatory cells in stroma}}{\text{Total intratumoral stromal area}} \times 100 \quad (2-1)$$

Only lymphocytes and plasma cells are included; polymorphonuclear leukocytes are excluded. Areas of necrosis, crush artifact, fibrosis, and hyalinization at the tumor

border are also excluded from the assessment. The score is reported as a single average percentage per tumor on H&E-stained full-tissue sections.

Clinically, tumors with sTILs $\geq 50\%$ are classified as lymphocyte-predominant breast cancer (LPBC), indicating a strong immune response. Additional commonly used thresholds include sTILs $< 10\%$ (TILs-low) and sTILs $\geq 30\%$ (TILs-high in many triple-negative breast cancer studies).

This scoring methodology is fundamentally an *area-based* measurement: it requires accurately delineating the boundary of each lymphocyte to compute its area contribution, not merely detecting its presence or counting its centroid. This distinction has direct implications for the design of automated nuclei analysis systems. Object detection models that output only bounding boxes or centroids are insufficient for sTILs scoring because they discard boundary geometry. Accurate sTILs computation requires methods that can both detect individual cells and delineate their boundaries, particularly in densely clumped regions where cell boundaries are ambiguous.

2.2.1.4 Cell Count-based Measurements

An alternative approach to quantifying immune infiltration is through cell count-based measurements, which estimate the number of immune cells per unit area rather than computing their proportional area. The most prominent example is the **Immunoscore**, developed and validated by Pagès et al. [15] for colorectal cancer. The Immunoscore quantifies the density of CD3⁺ and CD8⁺ T lymphocytes within two histological regions: the **tumor core** (CT) and the **invasive margin** (IM). This yields four density values, expressed as the number of positive cells per square millimeter:

$$\text{IS} = \{d_{\text{CD3}}^{\text{CT}}, d_{\text{CD3}}^{\text{IM}}, d_{\text{CD8}}^{\text{CT}}, d_{\text{CD8}}^{\text{IM}}\} \quad (2-2)$$

where d_m^r denotes the density of marker $m \in \{\text{CD3}, \text{CD8}\}$ in region $r \in \{\text{CT}, \text{IM}\}$. Each density value is classified as high (1) or low (0) based on a predefined percentile threshold determined from a reference patient cohort. The four binary values are then summed to produce the final Immunoscore:

$$\text{Immunoscore} = \sum_{m \in \{\text{CD3}, \text{CD8}\}} \sum_{r \in \{\text{CT}, \text{IM}\}} \mathbb{I}[d_m^r \geq \tau_m^r] \quad (2-3)$$

where τ_m^r is the density threshold for marker m in region r and $\mathbb{I}[\cdot]$ is the indicator function. The resulting score ranges from I0 (all four densities low) to I4 (all four densities high), with higher scores indicating stronger immune infiltration and consistently associated with improved patient survival across multiple validation cohorts.

Unlike area-based TILs scoring, count-based measurements do not require accurate boundary delineation of each individual cell. A detection model that correctly identifies cell centroids is sufficient to count cells per unit area. However, count-based methods typically require immunohistochemical (IHC) staining for specific markers (e.g., CD3, CD8) rather than standard H&E staining, which limits their applicability in resource-constrained settings where IHC is not routinely available. Furthermore, the area-based sTILs methodology has been more widely adopted across cancer types, particularly in breast cancer, making it the more broadly relevant measurement for automated analysis systems.

2.2.2 Digital Pathology Imaging

2.2.2.1 Tissue Slicing and Scanning

The digitization of histopathology slides produces whole-slide images (WSIs) through specialized scanners that capture high-resolution images of the entire tissue section. Modern slide scanners, such as those from Hamamatsu, Leica, and 3DHistech, use line-sensor or area-sensor cameras mounted on motorized stages to systematically capture overlapping tiles that are subsequently stitched into a single large image.

WSIs are typically stored in multi-resolution pyramidal formats (e.g., Aperio SVS, Hamamatsu NDP, or generic TIFF), where each level provides a different magnification. The spatial resolution of a WSI is specified in microns per pixel (MPP), which depends on the objective lens magnification used during scanning. Common resolutions range from approximately 0.25 MPP (equivalent to $40\times$ magnification) to 0.5 MPP ($20\times$). At 0.25 MPP, a typical biopsy slide can produce an image exceeding $50,000 \times 50,000$ pixels, necessitating patch-based processing strategies for computational analysis.

2.2.2.2 Histopathological Slides Staining

Tissue staining is essential because most biological tissues are nearly transparent under standard brightfield microscopy. Stains introduce contrast by selectively binding to tissue components through chemical interactions. The most widely used staining protocol in histopathology is Hematoxylin and Eosin (H&E) staining.

Hematoxylin is a basic (positively charged) dye that binds to acidic structures such as DNA and RNA, coloring nuclei dark blue or purple. Eosin is an acidic (negatively charged) dye that binds to basic structures such as cytoplasmic proteins and extracellular matrix, coloring them pink. The combination produces the characteristic appearance of H&E-stained tissue: blue-purple nuclei against a pink cytoplasmic and stromal background.

Beyond H&E, several specialized stains are used in clinical practice: Periodic

Acid-Schiff (PAS) for basement membranes and glycogen, Masson's Trichrome for collagen fibers, and immunohistochemical (IHC) stains for specific protein markers. Each stain produces distinct color characteristics that can be mathematically decomposed for computational analysis.

2.2.2.3 Beer-Lambert Law and Optical Density

The interaction of light with stained tissue follows the Beer-Lambert law, which describes the relationship between light absorption and the concentration of absorbing material. For a stained tissue sample, the transmitted intensity I at a given wavelength is related to the incident intensity I_0 by:

$$I = I_0 \cdot e^{-\alpha \cdot c \cdot d} \quad (2-4)$$

where α is the molar absorptivity (extinction coefficient), c is the concentration of the stain, and d is the path length through the tissue. The optical density (OD) is defined as:

$$\text{OD} = -\log_{10} \left(\frac{I}{I_0} \right) = \alpha \cdot c \cdot d \cdot \log_{10}(e) \quad (2-5)$$

In the context of digital pathology, the RGB pixel values of a scanned image are proportional to transmitted light intensity. Converting from RGB to optical density space is essential because OD is linearly additive: if multiple stains are present, the total OD at each pixel equals the sum of the OD contributions from each individual stain. This linearity property underpins color deconvolution methods.

2.2.2.4 Stain Vectors

A stain vector represents the characteristic color signature of a specific stain in optical density space. For an RGB image, each stain is described by a 3-element unit vector $\mathbf{s} = [s_R, s_G, s_B]^T$ in OD space, where each component indicates the stain's relative absorption at the red, green, and blue channels.

For H&E staining, the two stain vectors are:

- **Hematoxylin vector $\mathbf{s}_H$:** characterizes the absorption pattern of hematoxylin, which strongly absorbs in the blue-green range, producing its characteristic dark blue appearance.
- **Eosin vector $\mathbf{s}_E$:** characterizes the absorption pattern of eosin, which absorbs primarily in the green range, producing its pink appearance.

When two stains are present, the stain matrix S is formed by concatenating the

stain vectors as columns: $S = [s_H \ s_E]$. For three-stain protocols, a third column is added.

2.2.2.5 Stain Vector Estimation Methods

Since stain appearance varies across scanners, laboratories, and tissue types, the stain vectors must be estimated from each individual image or dataset. Several methods exist for automatic stain vector estimation:

Macenko's method [16] estimates stain vectors by analyzing the angular distribution of pixel OD values through Singular Value Decomposition (SVD). The procedure operates as follows.

Given an RGB image with N pixels, each pixel $\mathbf{p}_i = [R_i, G_i, B_i]^T$ is first converted to optical density space:

$$\mathbf{o}_i = -\log_{10} \left(\frac{\mathbf{p}_i}{255} + \epsilon \right) \quad (2-6)$$

where ϵ is a small constant to avoid logarithm of zero. Pixels whose OD magnitude $\|\mathbf{o}_i\|$ falls below a threshold t_{OD} are discarded, as they correspond to near-white background regions with negligible stain absorption. The threshold is typically set at the 1st percentile of the OD magnitude distribution.

For the remaining M pixels, a matrix $\hat{O} \in \mathbb{R}^{3 \times M}$ is formed by arranging the OD vectors as columns. SVD is then applied:

$$\hat{O} = U \Sigma V^T \quad (2-7)$$

where $U \in \mathbb{R}^{3 \times 3}$ contains the left singular vectors. The two principal singular vectors $\mathbf{u}_1$ and $\mathbf{u}_2$ (the first two columns of U) span the plane in which the stain vectors lie. Each OD vector is projected onto this plane:

$$\mathbf{p}'_i = \begin{bmatrix} \mathbf{u}_1^T \mathbf{o}_i \\ \mathbf{u}_2^T \mathbf{o}_i \end{bmatrix} \quad (2-8)$$

and converted to angular coordinates $\theta_i = \arctan 2(p'_{i,2}, p'_{i,1})$. The stain directions are determined by finding the angles $\theta_{\min}$ and $\theta_{\max}$ that enclose the 1st and 99th percentiles of the angular distribution, corresponding to the extreme concentrations of each stain.

The raw stain vectors are then reconstructed in OD space:

$$\mathbf{v}_1 = \mathbf{u}_1 \cos \theta_{\min} + \mathbf{u}_2 \sin \theta_{\min}, \quad \mathbf{v}_2 = \mathbf{u}_1 \cos \theta_{\max} + \mathbf{u}_2 \sin \theta_{\max} \quad (2-9)$$

Since SVD produces vectors with arbitrary sign, the vectors are flipped such that all components are non-negative (as OD values are inherently non-negative). Finally, the

vectors are normalized to unit length: $\mathbf{s}_j = \mathbf{v}_j / \|\mathbf{v}_j\|$.

To assign identities, the angle each normalized vector makes with a predefined reference direction in OD space (typically corresponding to a hematoxylin-like hue) is computed. The vector closer to the hematoxylin reference is labeled $\mathbf{s}_H$, and the other is labeled $\mathbf{s}_E$.

A residual stain vector is then computed via the cross product of the two estimated stain vectors:

$$\mathbf{s}_R = \mathbf{s}_H \times \mathbf{s}_E \quad (2-10)$$

This third vector is orthogonal to both $\mathbf{s}_H$ and $\mathbf{s}_E$ and captures the residual OD component not explained by the two primary stains, arising from background variation, optical noise, or imaging artifacts. It is included as the third column of the stain matrix to form a complete basis $S = [\mathbf{s}_H \ \mathbf{s}_E \ \mathbf{s}_R] \in \mathbb{R}^{3 \times 3}$, which is required for the matrix inversion in color deconvolution (Section??). Without this third vector, the stain matrix would be rank-deficient and non-invertible.

Macenko's method is widely adopted due to its computational simplicity and robustness, as the SVD-based projection efficiently captures the dominant stain directions without iterative optimization.

2.2.2.6 Color Deconvolution

Color deconvolution is the process of separating an H&E-stained image into its individual stain channels. Given the stain matrix S and the OD image $\mathbf{v}$ (a pixel vector in OD space), the stain concentrations $\mathbf{c}$ for each pixel are obtained by solving the linear system:

$$\mathbf{v} = S \cdot \mathbf{c} \quad (2-11)$$

Since S is a 3×2 matrix (for H&E) or 3×3 (for three stains), the system can be solved using the pseudoinverse:

$$\mathbf{c} = S^{-1} \cdot \mathbf{v} \quad (2-12)$$

where S^{-1} denotes the pseudoinverse (or inverse, for square S). The resulting stain concentration maps $\mathbf{c}$ provide separate grayscale images representing the hematoxylin and eosin density at each pixel. The hematoxylin channel is particularly useful for nuclei detection algorithms, as it isolates nuclear staining from cytoplasmic and stromal background.

2.2.2.7 Stain Normalization

Stain normalization addresses the significant color variation that exists across WSIs captured from different scanners, hospitals, or staining batches. This variation, if uncorrected, degrades the performance of deep learning models trained on data from one source and applied to another.

The standard approach to stain normalization involves three steps: (1) estimate the stain matrix S_{source} from the source image using one of the methods described above, (2) compute the stain concentration map $\mathbf{c} = S_{\text{source}}^{-1} \cdot \mathbf{v}_{\text{source}}$, and (3) reconstruct the normalized image using a target stain matrix S_{target} : $\mathbf{v}_{\text{normalized}} = S_{\text{target}} \cdot \mathbf{c}$, then convert back from OD to RGB space.

This process preserves the structural content of the image while standardizing color appearance. Stain normalization is commonly applied as a preprocessing step before training or inference in computational pathology pipelines.

2.2.3 Segmentation Paradigms

2.2.3.1 Semantic Segmentation

Semantic segmentation assigns a class label to every pixel in an image, producing a dense classification map where each pixel is labeled as belonging to one of several classes. Formally, given an input image $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$, a semantic segmentation model produces a label map $\mathbf{Y} \in \{1, \dots, K\}^{H \times W}$ where K is the number of classes and each $Y_{i,j}$ indicates the class of pixel (i, j) .

The standard architecture for semantic segmentation is the fully convolutional network (FCN), which replaces the fully connected layers of a classification network with convolutional layers to produce spatially dense outputs. The encoder-decoder structure is common: an encoder backbone (e.g., ResNet, VGG) progressively downsamples the input to extract high-level features, while a decoder progressively upsamples these features to recover the original spatial resolution. Skip connections between corresponding encoder and decoder layers help preserve fine-grained spatial details that would otherwise be lost during downsampling.

Semantic segmentation is effective for macro-level area measurements, such as estimating the overall proportion of stromal tissue occupied by lymphocytes. However, it does not distinguish between individual instances of the same class: all nuclei receive the same label regardless of their spatial separation, making it impossible to count individual cells.

2.2.3.2 Instance Segmentation

Instance segmentation extends semantic segmentation by assigning a unique label to each individual object instance, producing both a class label and an instance identity for every pixel. Given an input image, the output is a label map where each pixel is assigned a pair (c, k) indicating that it belongs to instance k of class c .

Two dominant paradigms exist for instance segmentation. **Proposal-based methods** such as Mask R-CNN [3] operate in two stages: a region proposal network (RPN) generates candidate object bounding boxes, and a mask head then predicts a binary segmentation mask for each proposal. While accurate, the two-stage pipeline introduces significant computational overhead, particularly when the number of instances is large.

Dense prediction methods such as U-Net [2] and HoVer-Net [7] predict instance information directly from dense feature maps. U-Net produces pixel-level embeddings where each pixel is mapped to a high-dimensional space, and instances are separated by clustering pixels based on embedding distance. HoVer-Net predicts horizontal and vertical distance maps from each pixel to the centroids of neighboring cells, using these maps to separate touching or overlapping instances. While these methods can achieve high accuracy, they require computationally expensive post-processing (e.g., watershed algorithms, clustering, or morphological operations) to convert dense predictions into individual instance masks.

2.2.3.3 Limitations of Segmentation for Nuclei Analysis

Segmentation-based approaches face two fundamental limitations when applied to nuclei analysis in histopathology images.

The **overlap problem** arises from the mutually exclusive pixel assignment inherent in segmentation. In a two-dimensional image, each pixel can only belong to one instance. When nuclei overlap in 3D tissue and are projected onto a 2D plane, their boundaries become entangled: a single pixel at the boundary between two overlapping nuclei cannot simultaneously represent both cells. Segmentation models resolve this ambiguity by drawing an artificial boundary between the two cells, which distorts the true morphology of both. This distortion is particularly problematic for nuclei analysis, where accurate boundary delineation directly affects area-based measurements such as TILs scoring.

The **computational bottleneck** stems from the dense prediction paradigm: segmentation models must classify every pixel in the image, including the vast majority of background pixels that carry no useful information. For a typical histopathology patch of 256×256 pixels, this means producing 65,536 individual predictions, even when only a small fraction correspond to actual nuclei. This per-pixel cost becomes prohibitive when processing whole-slide images, which may contain billions of pixels. The post-processing

required to extract individual instances from dense predictions (clustering, watershed, morphological operations) adds further computational overhead.

2.2.4 Detection Paradigms

Object detection models operate by dividing the input image into a spatial grid and predicting a fixed set of parameters (typically a bounding box and a class label) at each spatial location. Unlike segmentation, which classifies every pixel, detection produces a sparse set of structured outputs that describe the position and extent of individual objects. This grid-based prediction framework forms the foundation of modern single-stage detectors such as YOLO [5] and two-stage detectors such as Faster R-CNN [4], and is the paradigm adopted in this work.

2.2.4.1 Anchor Grid

In grid-based detection frameworks, predictions are organized around an **anchor grid**: a regular lattice of reference boxes (anchors) placed at each spatial location of a feature map. At each grid cell, multiple anchors of different scales and aspect ratios are defined to accommodate objects of varying sizes and shapes. Each anchor serves as a prior that the network refines into a final prediction by regressing coordinate offsets relative to the anchor dimensions and classifying the object category.

The anchor grid determines several key properties of the detector. It defines the maximum number of detections the model can produce: for a feature map of resolution $H \times W$ with A anchors per cell, the model generates up to $H \times W \times A$ predictions. Each grid cell is responsible for detecting objects whose centroids fall within its receptive field, creating a natural spatial correspondence between image regions and model outputs. This regular grid structure enables the detector to process objects at multiple scales simultaneously by operating on feature maps at different resolutions, with coarser grids detecting larger objects and finer grids detecting smaller objects.

2.2.4.2 Matching Mechanisms in Object Detection

A fundamental challenge in training detection models is establishing the correspondence between predicted outputs and ground-truth objects, a process known as **matching**. Given the anchor grid produces a large number of predictions per image, the matching mechanism determines which predictions are assigned to which ground-truth objects during loss computation.

During training, predictions are matched to ground-truth objects using an assignment rule. The assignment is formulated by constructing a **cost matrix** (also called a score matrix, depending on whether lower or higher values indicate better matches) that quantifies the quality of each prediction–ground-truth pair. Given N ground-truth objects

and M predictions, the cost matrix $\mathbf{C} \in \mathbb{R}^{N \times M}$ is defined such that element $C_{i,j}$ measures the dissimilarity between ground-truth object i and prediction j . The most common choice for $C_{i,j}$ is the negative IoU:

$$C_{i,j} = 1 - \text{IoU}(g_i, p_j) \quad (2-13)$$

where g_i is the bounding box (or polygon) of ground-truth object i and p_j is the predicted bounding box (or polygon). Since IoU ranges from 0 to 1, the cost $C_{i,j}$ also ranges from 0 (perfect overlap) to 1 (no overlap). Alternatively, some frameworks use a score matrix $\mathbf{S} \in \mathbb{R}^{N \times M}$ where $S_{i,j} = \text{IoU}(g_i, p_j)$ and the goal is to maximize the total score rather than minimize the total cost. The two formulations are equivalent: minimizing $\sum C_{i,j}$ is the same as maximizing $\sum S_{i,j}$ when $C_{i,j} = 1 - S_{i,j}$.

In addition to localization cost, the cost matrix may incorporate a **classification cost** term that penalizes mismatched class predictions. For multi-class detection with K classes, the classification cost can be defined using the cross-entropy between the predicted class distribution and the ground-truth class:

$$C_{i,j}^{\text{cls}} = -\log \hat{p}_j(c_i) \quad (2-14)$$

where $\hat{p}_j(c_i)$ is the predicted probability of class c_i (the true class of ground-truth i) and the total cost combines both terms:

$$C_{i,j} = C_{i,j}^{\text{loc}} + \lambda_{\text{cls}} \cdot C_{i,j}^{\text{cls}} \quad (2-15)$$

with λ_{cls} controlling the relative weight of classification versus localization. This combined cost matrix encodes both spatial alignment and semantic correctness into a single assignment objective.

Once the cost matrix is constructed, an assignment algorithm operates on it to determine which predictions correspond to which ground-truth objects. The choice of assignment algorithm determines whether the matching is one-to-many (allowing multiple predictions per ground-truth object), many-to-one (allowing one prediction to be matched to multiple ground-truth objects), or one-to-one (bipartite). Traditional methods such as Faster R-CNN [4] use IoU-based thresholding on this matrix: a prediction is assigned to a ground-truth object if $C_{i,j}$ falls below a threshold. This assignment is often ambiguous when multiple predictions overlap with the same ground-truth object, requiring heuristics to resolve conflicts. Modern YOLO variants use more sophisticated strategies, such as task-aligned matching and dual-label assignment, which extend the cost matrix with additional alignment terms. These strategies are discussed in Section 2.2.8.

2.2.4.3 Feature Pyramid Networks

A single feature map at a fixed resolution presents a fundamental limitation for object detection: objects in natural images vary widely in scale, and features that are discriminative for large objects (derived from deep, low-resolution layers with large receptive fields) may be too coarse to localize small objects, while features suitable for small objects (from shallow, high-resolution layers) may lack the semantic context needed to recognize large objects. **Feature Pyramid Networks (FPN)** [17] address this by constructing a multi-scale feature pyramid from a single input image in a computationally efficient manner.

An FPN extends a standard convolutional backbone with a top-down pathway and lateral connections. The backbone computes a hierarchy of feature maps $\{C_2, C_3, C_4, C_5\}$ at progressively coarser resolutions (downsampled by factors of 4, 8, 16, and 32 relative to the input). The top-down pathway then upsamples the deepest feature map C_5 and merges it with the corresponding lateral feature map from the backbone (e.g., C_4) via element-wise addition after applying a 1×1 convolution to match channel dimensions. This process is repeated to produce the pyramid levels $\{P_2, P_3, P_4, P_5\}$:

$$P_\ell = \text{Conv}_{1 \times 1}(C_\ell) + \text{Upsample}(P_{\ell+1}) \quad (2-16)$$

Each pyramid level P_ℓ has the same spatial resolution as its corresponding backbone feature map C_ℓ but benefits from the semantic richness of higher-level features propagated downward through the top-down pathway.

In a detector, each pyramid level is assigned to objects within a specific scale range. Large objects are detected at coarser levels (P_5) where the large receptive field captures holistic shape, while small objects are detected at finer levels (P_2) where high spatial resolution preserves precise localization. This per-level assignment reduces ambiguity in the matching process: objects of clearly different scales are assigned to different pyramid levels, preventing them from competing for the same prediction slot at a single resolution.

However, FPN also contributes to duplicate predictions. When an object falls near the boundary between two assigned scale ranges, it may be detected at two adjacent pyramid levels simultaneously, producing two spatially overlapping predictions for the same object. These cross-level duplicates are a significant source of redundancy that downstream NMS must resolve.

2.2.4.4 Non-Maximum Suppression

Non-maximum suppression (NMS) is a post-processing step used by most detection frameworks to eliminate duplicate predictions. When multiple predictions overlap with

the same ground-truth object, NMS selects the prediction with the highest confidence score and suppresses (removes) all other predictions whose overlap with the selected one exceeds a predefined IoU threshold.

The standard NMS procedure operates as follows. Given a set of detections ranked by confidence score, the highest-confidence detection is selected and all remaining detections with IoU exceeding a threshold τ_{nms} are suppressed. This process repeats until no detections remain.

Formally, let $\mathcal{D} = \{d_1, d_2, \dots, d_N\}$ be a set of detections sorted in descending order of confidence s_i , each with bounding box B_i . The NMS-filtered output set $\mathcal{D}'$ is defined inductively:

$$\mathcal{D}' = \{d_i \in \mathcal{D} : \forall d_j \in \mathcal{D}' \text{ with } s_j > s_i, \text{IoU}(B_i, B_j) < \tau_{\text{nms}}\} \quad (2-17)$$

Alternatively, expressed as a greedy iteration that produces a maximal independent set with respect to the τ_{nms} -thresholded neighborhood graph: starting from $\mathcal{D}' = \emptyset$, for each $d_i \in \mathcal{D}$ in confidence order, if $\max_{d_j \in \mathcal{D}'} \text{IoU}(B_i, B_j) < \tau_{\text{nms}}$, then $\mathcal{D}' \leftarrow \mathcal{D}' \cup \{d_i\}$.

The IoU threshold τ_{nms} controls the trade-off between eliminating duplicates and preserving valid detections of closely spaced objects.

NMS introduces a fundamental tension in nuclei detection. In densely clumped regions, multiple nuclei overlap significantly, producing predictions that are spatially close and have high mutual IoU. A low NMS threshold aggressively suppresses these predictions, reducing duplicates but also removing valid detections of distinct cells. A high threshold preserves more predictions but allows duplicates to remain, inflating the cell count. Neither setting resolves the underlying problem: NMS cannot distinguish between two overlapping predictions of the same cell and two valid predictions of adjacent cells that happen to overlap.

2.2.4.5 Limitations of Detection for Nuclei Analysis

Detection models offer computational advantages over segmentation but face two critical shortcomings when applied to nuclei analysis.

The **boundary problem** is that bounding box representations discard cell morphology. A bounding box encloses the approximate spatial extent of a cell but does not capture its shape. For approximately circular nuclei, the difference between bounding box area and true cell area may be modest, but for irregularly shaped or overlapping nuclei, bounding boxes significantly overestimate cell size and destroy the geometric information necessary for area-based measurements such as TILs scoring. A detection model can identify where cells are but cannot determine how much area each cell occupies.

The **density problem** compounds the boundary problem in the very regions where accuracy matters most. In densely clumped areas, cells are closely spaced and their bounding boxes overlap heavily. NMS must then decide which overlapping boxes correspond to distinct cells and which are duplicate predictions of the same cell. Since this decision is based solely on geometric overlap and confidence scores, NMS systematically undercounts in clumped regions: valid detections of adjacent cells are suppressed because their boxes overlap, while in sparse regions where cells are well-separated, NMS has little effect. This creates a density-dependent counting bias where the model performs worst precisely in the regions most relevant to clinical assessment.

2.2.5 Convolutional Neural Networks

2.2.5.1 Convolution and Feature Extraction

A convolutional layer applies a set of learnable filters (kernels) to an input feature map, producing an output feature map where each element is a weighted sum of local input elements. For a 2D convolution with input $\mathbf{X} \in \mathbb{R}^{H \times W \times C_{\text{in}}}$ and kernel $\mathbf{W} \in \mathbb{R}^{K \times K \times C_{\text{in}} \times C_{\text{out}}}$, the output at spatial position (i, j) for channel c is:

$$\mathbf{Y}_{i,j,c} = \sum_{m=0}^{K-1} \sum_{n=0}^{K-1} \sum_{k=0}^{C_{\text{in}}-1} \mathbf{W}_{m,n,k,c} \cdot \mathbf{X}_{i+m,j+n,k} + b_c \quad (2-18)$$

where b_c is the bias term for output channel c . Each kernel slide across the input computes the same operation at every spatial position, exploiting the translation invariance of visual patterns. Stacking multiple convolutional layers with nonlinear activation functions (typically ReLU) enables the network to learn hierarchical feature representations: early layers capture low-level features such as edges and textures, while deeper layers capture higher-level structures such as cell shapes and tissue patterns.

2.2.5.2 Receptive Field and Kernel Size

The receptive field of a neuron in a convolutional network is the region of the input image that influences its output. For a single convolutional layer with kernel size K , each neuron has a receptive field of $K \times K$ pixels. Stacking L convolutional layers, each with kernel size K and stride 1, produces a receptive field of size $(K + (K - 1)(L - 1)) \times (K + (K - 1)(L - 1))$ at the deepest layer. Using stride $s > 1$ or pooling layers between convolutions further enlarges the effective receptive field.

The kernel size directly determines the model’s ability to capture spatial context. A 3×3 kernel sees only immediate neighbors, while a 7×7 kernel captures a larger local context. However, larger kernels increase the number of parameters quadratically: a 7×7 kernel has 5.4 times more parameters than a 3×3 kernel for the same number of channels.

Modern architectures typically use small kernels (3×3) stacked deeply to achieve large effective receptive fields while keeping the parameter count manageable.

For nuclei detection, a sufficiently large receptive field is critical because nuclei can vary significantly in size, and contextual information from surrounding tissue helps distinguish nuclei from artifacts and debris. Methods such as LKCell [18] demonstrate that using large convolution kernels (e.g., 7×7 or larger) with structural reparameterization can achieve large receptive fields with lower computational cost than stacking many small-kernel layers.

2.2.5.3 Depthwise Convolution

Depthwise convolution is a spatial decomposition of the standard convolution operation that significantly reduces computational cost. In a standard convolution with C_{in} input channels and C_{out} output channels using a $K \times K$ kernel, the number of multiplications per spatial position is $K^2 \cdot C_{\text{in}} \cdot C_{\text{out}}$. Depthwise convolution decomposes this into two steps:

1. **Depthwise step:** Each of the C_{in} input channels is convolved independently with its own $K \times K$ kernel, producing C_{in} intermediate feature maps. Cost: $K^2 \cdot C_{\text{in}}$ multiplications per position.
2. **Pointwise step:** A 1×1 convolution (pointwise convolution) then combines the C_{in} intermediate maps into C_{out} output channels. Cost: $C_{\text{in}} \cdot C_{\text{out}}$ multiplications per position.

The total cost is $K^2 \cdot C_{\text{in}} + C_{\text{in}} \cdot C_{\text{out}}$ per position, compared to $K^2 \cdot C_{\text{in}} \cdot C_{\text{out}}$ for standard convolution. The reduction factor is approximately K^2 , making depthwise convolution particularly effective for larger kernel sizes. This efficiency makes depthwise separable convolutions a key building block in lightweight architectures, as demonstrated by MobileNet [19] and adopted in subsequent YOLO variants for efficient feature extraction.

2.2.6 Discrete Wavelet Transform

2.2.6.1 Wavelet Decomposition Fundamentals

The wavelet transform provides a time-frequency (or space-frequency) analysis of a signal by decomposing it into components localized in both the spatial and frequency domains. Unlike the Fourier transform, which projects a signal onto infinite sinusoids and loses all spatial localization, the wavelet transform uses basis functions of finite support that capture both *where* and *at what scale* features occur.

The **Continuous Wavelet Transform (CWT)** of a signal $f(t) \in L^2(\mathbb{R})$ is defined

with respect to a *mother wavelet* $\psi(t)$ [20]:

$$W_f(a, b) = \frac{1}{\sqrt{|a|}} \int_{-\infty}^{\infty} f(t) \psi^* \left(\frac{t - b}{a} \right) dt \quad (2-19)$$

where $a \in \mathbb{R} \setminus \{0\}$ is the scale (dilation) parameter, $b \in \mathbb{R}$ is the translation (shift) parameter, and ψ^* denotes the complex conjugate. The factor $|a|^{-1/2}$ ensures energy normalization across scales. Small values of a produce narrow, high-frequency wavelets that capture fine details, while large values of a produce wide, low-frequency wavelets that capture coarse structures. By varying both a and b , the CWT produces a two-dimensional representation of the signal in the scale-translation plane, revealing how frequency content evolves across spatial position.

The CWT suffers from redundancy: the transform of a one-dimensional signal yields a two-dimensional coefficient map, and neighboring scales and translations are highly correlated. For computational efficiency and practical signal processing, the wavelet transform is commonly discretized.

2.2.6.2 Discrete Wavelet Transform

The **Discrete Wavelet Transform (DWT)** replaces the continuous scale-translation parameters with discrete samples, most commonly on a dyadic grid where $a = 2^{-j}$ and $b = k \cdot 2^{-j}$ for integers $j, k \in \mathbb{Z}$ [21]. Under a multiresolution analysis framework [21], the DWT can be efficiently implemented as a cascade of filter banks. At each level, the signal is convolved with a pair of quadrature mirror filters: a low-pass filter $h[n]$ associated with the scaling function $\phi(t)$, and a high-pass filter $g[n]$ associated with the wavelet function $\psi(t)$, followed by downsampling by a factor of 2:

$$c_{j+1}[k] = \sum_n h[n - 2k] c_j[n], \quad d_{j+1}[k] = \sum_n g[n - 2k] c_j[n] \quad (2-20)$$

where c_j are the approximation coefficients at level j and d_j are the detail coefficients. The approximation c_{j+1} captures the low-frequency content at coarser resolution, while the detail d_{j+1} captures the high-frequency information lost during downsampling.

For a 2D signal such as an image, the DWT is applied separably: first filtering rows, then filtering columns of the intermediate result. This produces four subbands per decomposition level:

- **LL (Low-Low):** Approximation coefficients, obtained by applying the low-pass filter h to both rows and columns. This subband is a downsampled version of the input that

preserves the overall spatial layout.

- **LH** (Low-High): Horizontal detail coefficients, obtained by applying h to rows and g to columns, capturing vertical edges.
- **HL** (High-Low): Vertical detail coefficients, obtained by applying g to rows and h to columns, capturing horizontal edges.
- **HH** (High-High): Diagonal detail coefficients, obtained by applying g to both rows and columns, capturing diagonal edges and fine textures.

Each subband has half the spatial resolution of the input in both dimensions. The decomposition can be applied recursively to the LL subband to obtain multi-level wavelet representations, where each level captures features at a different spatial scale.

The key advantage of wavelet decomposition over standard downsampling (e.g., strided convolution or max pooling) is that it provides a principled frequency separation. Standard downsampling discards high-frequency information entirely, which can eliminate fine textures and edges critical for accurate boundary delineation. Wavelet decomposition preserves high-frequency details in the LH, HL, and HH subbands, making these details available for subsequent processing.

2.2.6.3 Wavelet-Based Downsampling in CNNs

Integrating DWT into convolutional neural networks replaces standard downsampling operations (strided convolution or max pooling) with wavelet-based downsampling. In this approach, a feature map at one resolution is decomposed into its four wavelet subbands using DWT. The LL subband serves as the downsampled feature map for the next network stage, while the detail subbands (LH, HL, HH) can be concatenated with or injected into intermediate layers to preserve high-frequency information.

The wavelet-based downsampling operation can be expressed as follows. Given a feature map $\mathbf{F} \in \mathbb{R}^{H \times W \times C}$, the DWT produces four subbands each of size $\frac{H}{2} \times \frac{W}{2} \times C$. The LL subband $\mathbf{F}_{LL}$ is passed to the next convolutional block as the primary feature representation. The detail subbands can be processed by lightweight convolutional layers and merged back into the feature pipeline through skip connections or attention mechanisms.

This design is particularly relevant for nuclei detection, where cell boundaries are defined by fine high-frequency edges in the H&E image. By preserving these details through the downsampling hierarchy, wavelet-based networks can maintain boundary-aware features at deeper layers where the detection head operates. WaveConvX [22] demonstrates this principle in the context of histopathology image classification, showing that wavelet integration improves feature quality for FCN-based architectures.

2.2.7 Star-Convex Object Representation

2.2.7.1 Geometric Definitions

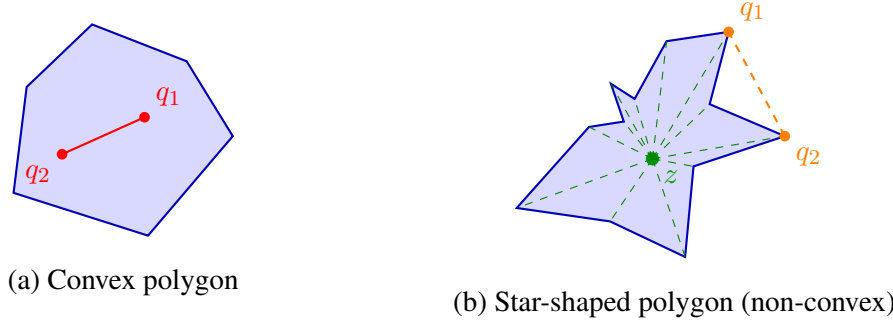


Figure 2.1. Illustration of convex and star-shaped polygons. (a) A convex polygon: the segment $\overline{q_1q_2}$ connecting any two interior points lies entirely within the polygon. (b) A star-shaped polygon that is not convex: the kernel point z (near the center) sees all boundary vertices (dashed green), yet the segment $\overline{q_1q_2}$ connecting two interior points exits the polygon, demonstrating that the interior is not a convex set.

A **convex polygon** is a polygon whose interior forms a convex set. A polygon P is convex if its interior region is a convex set [23]. A region R is a convex set if, for any two points q_1 and q_2 in R , the segment $\overline{q_1q_2}$ is entirely contained in R [23]. Intuitively, for a convex polygon, the line segment connecting any two points inside the polygon lies entirely within the polygon, and no interior angle exceeds 180° .

A **star-shaped polygon** generalizes the notion of convexity. A polygon P is star-shaped if there exists a point z not external to P such that for all points p of P , the line segment $\overline{zp}$ lies entirely within P [23]. The set of all such points z is called the *kernel* of the polygon. A star-shaped polygon can be viewed as one where there is at least one interior point visible to every other point in the polygon. Every convex polygon is star-shaped (the kernel is the entire interior), but a star-shaped polygon may have indentations that make it non-convex.

A **star-convex polygon** is a polygon P that is simultaneously star-shaped and convex. This dual property makes star-convex polygons particularly useful for representing cell nuclei: because they are convex, every boundary point can be uniquely described by a single ray emanating from the centroid, and because they are star-shaped, the centroid can be placed at a kernel point where all such rays intersect the boundary exactly once. In polar coordinates, a star-convex polygon centered at its centroid c can therefore be completely represented by a set of radial distances $\{d_0, d_1, \dots, d_{n-1}\}$ measured at n fixed angular intervals from the centroid to the boundary.

2.2.7.2 Suitability for Nuclei Representation

The star-convex polygon representation is well-suited for representing cell nuclei because of their characteristic morphology. The majority of cell nuclei encountered in H&E-stained tissue sections are approximately convex or roundish in shape [10]. Healthy epithelial nuclei are round or elliptical, fibroblasts and endothelial cells are elongated but convex, and even malignant nuclei with pleomorphic variations largely retain a convex silhouette in their two-dimensional projection. This near-ubiquitous convexity means that a star-convex polygon can capture the full boundary geometry of a nucleus with a compact set of radial distances, avoiding the information loss of axis-aligned bounding boxes while remaining far more parameter-efficient than dense per-pixel segmentation masks [6].

2.2.8 End-to-End Detection with Bipartite Matching

2.2.8.1 Bipartite Matching: Hungarian Algorithm

Bipartite matching establishes a one-to-one correspondence between two sets of elements by finding an optimal assignment that minimizes a total cost. In the context of object detection, bipartite matching assigns each ground-truth object to exactly one prediction and each prediction to at most one ground-truth object, eliminating the ambiguity inherent in IoU-based assignment where multiple predictions can match the same ground-truth object.

The assignment is formulated as a minimum-cost bipartite matching problem. Given a set of N ground-truth objects and M predictions, a cost matrix $\mathbf{C} \in \mathbb{R}^{N \times M}$ is constructed where $C_{i,j}$ measures the dissimilarity between ground-truth i and prediction j . A common choice is the negative IoU: $C_{i,j} = 1 - \text{IoU}(g_i, p_j)$. The optimal assignment is found by solving:

$$\begin{aligned}
& \min_{x_{i,j}} \sum_{i=1}^N \sum_{j=1}^M C_{i,j} x_{i,j} \\
& \text{s.t.} \quad \sum_{j=1}^M x_{i,j} = 1, \quad \forall i \in \{1, \dots, N\} \\
& \quad \quad \sum_{i=1}^N x_{i,j} \leq 1, \quad \forall j \in \{1, \dots, M\} \\
& \quad \quad x_{i,j} \in \{0, 1\}
\end{aligned} \tag{2-21}$$

where $x_{i,j} = 1$ if ground-truth i is assigned to prediction j , and 0 otherwise. The first constraint ensures every ground-truth object receives exactly one match. The second constraint ensures no prediction is matched to more than one ground-truth object. When $M > N$, predictions not assigned to any ground-truth are treated as background.

The **Hungarian algorithm** (Kuhn–Munkres) solves this problem in $O(\max(N, M)^3)$ time through the following steps on a square cost matrix (padded with large values if $N \neq M$): (1) subtract the row minimum from each row, (2) subtract the column minimum from each column, (3) cover all zero entries with the minimum number of horizontal and vertical lines, (4) if the number of lines equals N , an optimal assignment is found by selecting independent zeros; otherwise, subtract the smallest uncovered value from all uncovered entries, add it to all doubly covered entries, and repeat from step 3.

This bipartite matching formulation is the foundation of end-to-end detection frameworks such as DETR and LSP-DETR [8]. By guaranteeing a unique assignment, it eliminates the need for NMS as a post-processing step, since each ground-truth object is matched to exactly one prediction and there are no duplicate assignments to suppress.

2.2.8.2 Task-Aligned Matching

Task-aligned matching is an assignment strategy used in YOLO-based detection frameworks that jointly considers classification and localization quality when matching predictions to ground-truth objects. Rather than using IoU alone, task-aligned matching defines a combined alignment metric:

$$t = s^\alpha \cdot u^\beta \quad (2-22)$$

where s is the classification confidence score, u is the predicted IoU between the prediction and the ground-truth object, and α, β are weighting hyperparameters that balance the contribution of classification and localization. A prediction is assigned to a ground-truth object only if its alignment metric t exceeds a threshold, and each ground-truth object is assigned to the prediction with the highest t value. This ensures that predictions assigned to an object are both well-localized (high IoU) and confidently classified, improving training stability.

2.2.8.3 Dual-Label Assignment

Dual-label assignment, introduced in YOLO26, is a training strategy that combines one-to-many and one-to-one matching within a single architecture to achieve the benefits of both paradigms. The model employs two parallel detection heads: a **one-to-many head** that assigns multiple predictions to each ground-truth object (as in conventional YOLO training), and a **one-to-one head** that enforces a unique bipartite matching between predictions and ground-truth objects (as in DETR-style training).

The two heads are balanced through an **annealing weight schedule** that controls their relative contribution to the total loss. During early training epochs, the one-to-many head dominates, providing dense gradient signals that stabilize training and accelerate

feature learning. As training progresses, the weight shifts gradually toward the one-to-one head, which refines the predictions to produce unambiguous, duplicate-free detections. This transition allows the model to benefit from the richer supervision of one-to-many matching early on while converging to the cleaner one-to-one assignment necessary for NMS-free inference. At inference time, only the one-to-one head is used, eliminating the need for non-maximum suppression.

2.2.9 Evaluation Metrics

2.2.9.1 Precision, Recall, and F1 Score

Object detection and segmentation are evaluated by comparing predicted instances against ground-truth annotations. Each prediction is classified as a **true positive (TP)** if it correctly matches a ground-truth object, a **false positive (FP)** if it does not correspond to any ground-truth object, or a **false negative (FN)** if a ground-truth object is missed. The definition of a correct match depends on the task: for bounding-box or polygon-based detection, a prediction is typically considered a TP if its intersection over union (IoU) with the ground truth exceeds a threshold τ (commonly $\tau = 0.5$); for centroid-based detection, a prediction may be considered a TP if the distance between the predicted and ground-truth centroids falls below a distance threshold.

Given the counts of TP, FP, and FN, precision and recall are defined as:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \quad (2-23)$$

Precision measures the fraction of predictions that are correct (fewer FPs yield higher precision), while recall measures the fraction of ground-truth objects that are found (fewer FNs yield higher recall). The two are often in tension: a model that predicts conservatively achieves high precision but low recall, while a model that predicts aggressively achieves high recall but low precision.

The **F1 score** is the harmonic mean of precision and recall, providing a single balanced metric:

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (2-24)$$

2.2.9.2 Mean Average Precision (mAP)

Mean Average Precision (mAP) is the standard evaluation metric for object detection. It is computed by first calculating precision–recall curves for each object class and then averaging the area under these curves across all classes.

Given a set of detections ranked by confidence score and a chosen match criterion

(typically $\text{IoU} > 0.5$), precision and recall are computed at each confidence threshold to produce a precision–recall curve. The Average Precision (AP) for a single class is then the area under the interpolated precision–recall curve:

$$\text{AP} = \int_0^1 P_{\text{interp}}(R) dR \quad (2-25)$$

where $P_{\text{interp}}(R) = \max_{\tilde{R} \geq R} P(\tilde{R})$ is the interpolated precision at recall level R . In practice, this is approximated by evaluating precision at a discrete set of recall points.

The mean Average Precision is then the mean of per-class AP values:

$$\text{mAP} = \frac{1}{C} \sum_{c=1}^C \text{AP}_c \quad (2-26)$$

where C is the number of object classes. When reported at a specific IoU threshold (e.g., $\text{mAP}@0.5$), detections are matched using that threshold. The COCO evaluation protocol averages mAP over multiple IoU thresholds:

$$\text{mAP}@[.50:.05:.95] = \frac{1}{K} \sum_{k=1}^K \text{mAP}(\theta_k) \quad (2-27)$$

where $\theta_k \in \{0.50, 0.55, \dots, 0.95\}$ and $K = 10$. This provides a more comprehensive assessment of localization accuracy than evaluation at a single threshold. The notation $\text{mAP}@0.50$ (or AP_{50}) refers to evaluation at a single IoU threshold of 0.50.

2.2.9.3 Panoptic Quality (PQ)

Panoptic Quality (PQ) was introduced by Kirillov et al. [24] as a unified metric for panoptic segmentation, combining both recognition quality (detection accuracy) and segmentation quality (boundary accuracy). PQ has become the standard evaluation metric for nuclei instance segmentation benchmarks such as PanNuke [25] and CoNIC [26].

Given a ground truth segmentation G and a predicted segmentation P , each instance in G and P is uniquely labeled. A predicted segment $\hat{s}$ is matched to a ground truth segment s if and only if their intersection over union (IoU) exceeds a threshold, typically $\text{IoU} > 0.5$:

$$\text{IoU}(s, \hat{s}) = \frac{|s \cap \hat{s}|}{|s \cup \hat{s}|} \quad (2-28)$$

After matching, PQ is computed as:

$$\text{PQ} = \underbrace{\frac{|TP|}{|TP| + \frac{1}{2}|FP| + \frac{1}{2}|FN|}}_{\text{Recognition Quality (RQ)}} \times \underbrace{\frac{\sum_{(s,\hat{s}) \in TP} \text{IoU}(s, \hat{s})}{|TP|}}_{\text{Segmentation Quality (SQ)}} \quad (2-29)$$

where TP is the set of true positive pairs (matched ground truth–prediction pairs), FP is the set of false positive predictions (unmatched predictions), and FN is the set of false negative ground truth segments (unmatched ground truth). PQ thus decomposes into two interpretable components:

- **Recognition Quality (RQ):** measures detection accuracy and is equivalent to the F1 score computed over matched and unmatched segments.
- **Segmentation Quality (SQ):** measures the average IoU of correctly matched pairs, quantifying how precisely the predicted boundaries align with the ground truth.

Several PQ variants are commonly reported in nuclei segmentation literature:

- **Instance PQ (iPQ):** evaluates PQ on a per-image basis and then averages across all images in the dataset. Each image contributes equally regardless of the number of nuclei it contains.
- **Multi-class PQ (mPQ):** extends PQ to multi-class settings by computing PQ independently for each nuclei type and then averaging across all classes:

$$\text{mPQ} = \frac{1}{C} \sum_{c=1}^C \text{PQ}_c \quad (2-30)$$

where C is the number of classes and PQ_c is the PQ computed for class c only.

- **Binary PQ (bPQ):** computes PQ in a binary setting where all nuclei types are treated as a single class, ignoring type-specific distinctions. This is useful for evaluating pure instance segmentation performance independent of classification accuracy.
- **PanNuke PQ (nuPQ):** the specific PQ variant used in the PanNuke benchmark, which computes mPQ across the 19 nuclei type categories defined in the dataset.
- **Masked PQ (MPQ):** a variant introduced by LSP-DETR [8] designed for fair evaluation of overlapping nuclei. Standard PQ penalizes polygon predictions that cross artificially separated ground-truth boundaries in regions of biological overlap. MPQ replaces the standard IoU with a **masked IoU (mIoU)** that isolates evaluation to the matched ground-truth instance. Given a ground-truth mask m_k and predicted polygon $\hat{m}_k$, and letting $\mathcal{G} = \bigcup_{k=1}^K m_k$ be the union of all ground-truth masks:

$$\text{mIoU}(m_k, \hat{m}_k) = \frac{|m_k \cap \hat{m}_k|}{|m_k \cup (\hat{m}_k \setminus \mathcal{G})|} \quad (2-31)$$

$$\text{MPQ} = \underbrace{\frac{|TP|}{|TP| + \frac{1}{2}|FP| + \frac{1}{2}|FN|}}_{\text{RQ}} \times \underbrace{\frac{\sum_{(m_k, \hat{m}_k) \in TP} \text{mIoU}(m_k, \hat{m}_k)}{|TP|}}_{\text{SQ}_{\text{mask}}} \quad (2-32)$$

The denominator in mIoU includes the ground-truth area plus only those parts of the prediction that fall outside $\mathcal{G}$, meaning predictions overlapping *other* ground-truth masks beyond the matched one are penalized, but overlap with the matched instance itself is not double-counted. This makes MPQ the appropriate metric for polygon-based methods that output realistically overlapping cell representations.

2.2.9.4 Aggregated Jaccard Index (AJI)

The Aggregated Jaccard Index (AJI) [27] is an instance-level segmentation metric designed to address the overcounting problem in the standard Jaccard Index when applied to multi-object segmentation. In nuclei segmentation, the standard Jaccard Index (Intersection over Union, IoU) computed per instance and then averaged can overestimate performance because each ground-truth nucleus is free to match with any predicted nucleus independently, and unmatched predictions do not penalize the score.

AJI resolves this by computing a single global intersection-over-union across all instances. Given a set of ground-truth nuclei $G = \{g_1, g_2, \dots, g_N\}$ and predicted nuclei $P = \{p_1, p_2, \dots, p_M\}$, AJI first finds the optimal one-to-one assignment that maximizes total pairwise IoU. Let $\mathcal{M} \subseteq G \times P$ denote the set of matched pairs, $G_{\text{matched}} \subseteq G$ the set of matched ground-truth nuclei, and $P_{\text{matched}} \subseteq P$ the set of matched predicted nuclei. The AJI is then defined as:

$$\text{AJI} = \frac{\sum_{(g_i, p_j) \in \mathcal{M}} |g_i \cap p_j|}{\sum_{(g_i, p_j) \in \mathcal{M}} |g_i \cup p_j| + \sum_{g_k \in G \setminus G_{\text{matched}}} |g_k| + \sum_{p_l \in P \setminus P_{\text{matched}}} |p_l|} \quad (2-33)$$

The numerator sums the intersection area over all matched pairs, while the denominator sums the union area over all matched pairs plus the areas of all unmatched ground-truth and predicted nuclei. This formulation penalizes both false negatives (unmatched ground-truth nuclei) and false positives (unmatched predictions), providing a more stringent evaluation than per-instance averaging.

CHAPTER III

RESEARCH METHODOLOGY

3.1 Datasets

This study used publicly available secondary datasets for training and evaluation. Table 4.1 summarized the datasets and their roles.

Table 3.1. Summary of datasets used in this study.

Dataset	Purpose	Tissue	Format
PanNuke	Primary benchmarking	Multi-organ	Pickled bytes
PUMA	TILs detection	Melanoma	TIFF, GeoJSON
panopTILs	TILs detection	Breast	PNG
MoNuSAC	TILs detection	Multi-organ	Pickled bytes

3.1.1 PanNuke

PanNuke [25, 28] is a large-scale pan-cancer nucleus instance segmentation dataset containing approximately 189,744 labeled nuclei across 19 tissue types. Each nucleus is annotated with an instance mask and one of five nuclear categories. The dataset is provided as pickled bytes containing image patches and corresponding label masks. In this study, PanNuke served as the primary benchmarking dataset for evaluating nuclei detection and segmentation performance.

3.1.2 PUMA

PUMA (Melanoma Histopathology Dataset with Tissue and Nuclei Annotations) [29] provides annotated histopathology images for TILs detection in melanoma tissue. The dataset is distributed as TIFF images paired with GeoJSON annotation files containing polygon-level cell boundary annotations. This dataset was used to evaluate RayCastED's ability to detect and delineate TILs in a clinically relevant setting.

3.1.3 panopTILs

panopTILs [30] provides TILs annotations on breast cancer histopathology images. The dataset is distributed as PNG image patches with corresponding annotation files. This dataset supplemented the TILs evaluation alongside PUMA, providing breast tissue samples for assessing TILs detection performance across cancer types.

3.1.4 MoNuSAC

MoNuSAC [31] provides annotated nuclei across multiple organ types with four nuclear categories: epithelial, lymphocyte, neutrophil, and macrophage. The dataset is provided as pickled bytes. The lymphocyte annotations in MoNuSAC were particularly relevant for evaluating TILs detection, as lymphocytes are the primary cell type of interest in TILs scoring.

3.2 Methods

3.2.1 Data Preprocessing

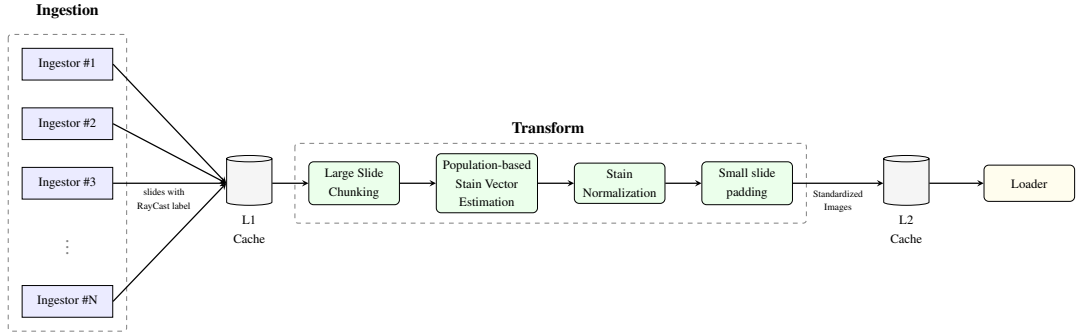


Figure 3.2. High-level Pipeline Design

3.2.1.1 Ingestion Pipeline

RayCast Representation Generation.

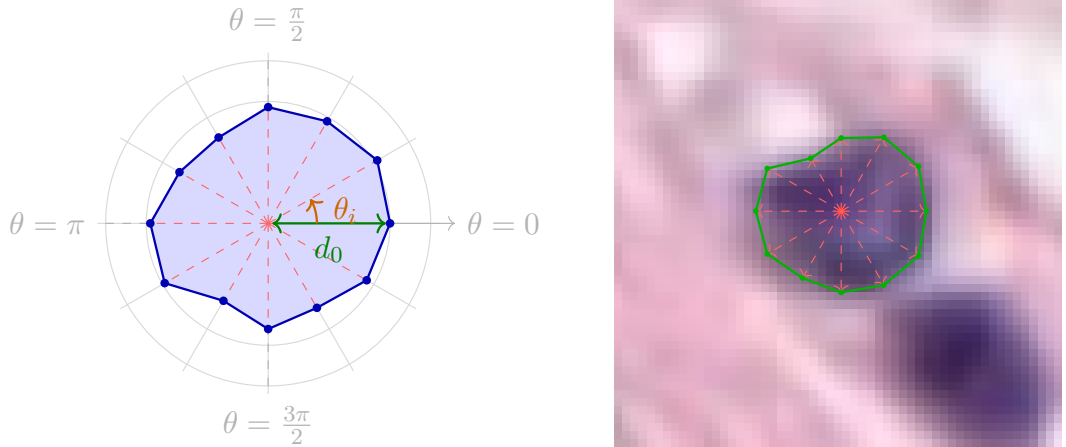


Figure 3.3. Star-convex polygon property. (Left) A star-convex polygon with centroid c , polar grid, and 12 radial rays with angular spacing θ_i . (Right) The same representation overlaid on a histopathology cell image.

Given a closed polygon $\mathcal{P}$ defined by an ordered set of K vertices $\{\mathbf{V}_j\}_{j=0}^{K-1}$ in $\mathbb{R}^2$ and a class label c , the RayCast representation encodes the polygon as a fixed-length

vector by expressing its boundary as radial distances from a chosen centroid along R uniformly spaced angular directions. The output is an annotation vector:

$$\mathbf{a} = (c, c_x, c_y, d_0, d_1, \dots, d_{R-1})^\top \in \mathbb{R}^{3+R} \quad (3-1)$$

where (c_x, c_y) is the centroid, R is the number of rays (set to 32 in this study, and constrained to be a multiple of 4), and d_r records the Euclidean distance from the centroid to the polygon boundary along ray r at angle $\theta_r = r \cdot 2\pi/R$. The convention is $\theta_0 = 0$ (east, $+X$ axis), increasing counter-clockwise. Figure 3.4 illustrates the concept.

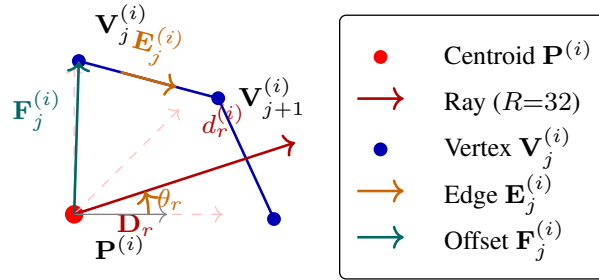


Figure 3.4. RayCast representation of a cell polygon. From the area-weighted centroid P , $R = 32$ rays are cast at uniformly spaced angles. Each distance d_r records the nearest boundary intersection along ray r .

The generation algorithm proceeds in four stages: (1) centroid computation, (2) containment verification and exterior centroid correction, (3) ray–edge intersection solving, and (4) nearest-hit selection and output assembly. The entire pipeline is formulated as batched tensor operations for GPU parallelism, processing all polygons in a region of interest simultaneously.

Raster Mask to Contour Extraction.

Datasets such as PanNuke and MoNuSAC provide per-cell binary raster masks rather than polygon coordinates. For each cell, the binary mask is decoded and contour vertices are extracted via Moore-neighbor tracing (OpenCV’s `findContours` with `CHAIN_APPROX_NONE`) to preserve full boundary fidelity. When a mask produces multiple contours, only the contour with the largest enclosed area is retained. Contours with fewer than 3 vertices are discarded.

Centroid Computation.

The area-weighted centroid serves as the origin from which all rays emanate. Given the polygon vertices with cyclic indexing $V_K \equiv V_0$, the signed area is computed via the shoelace formula:

$$A = \frac{1}{2} \sum_{j=0}^{K-1} (x_j y_{j+1} - x_{j+1} y_j) \quad (3-2)$$

Polygons with $|A| < \varepsilon$ (where $\varepsilon = 10^{-12}$) are near-zero area; in this case the centroid is set to the first vertex $\mathbf{V}_0$. The area-weighted centroid coordinates are:

$$c_x = \frac{1}{6A} \sum_{j=0}^{K-1} (x_j + x_{j+1}) (x_j y_{j+1} - x_{j+1} y_j) \quad (3-3)$$

$$c_y = \frac{1}{6A} \sum_{j=0}^{K-1} (y_j + y_{j+1}) (x_j y_{j+1} - x_{j+1} y_j) \quad (3-4)$$

For GPU-batched processing across N polygons padded to $K_{\max} = \max_i K_i$ vertices, an edge mask $\mathbf{M}^{\text{edge}} \in \{0, 1\}^{N \times K_{\max}}$ distinguishes real polygon edges from padding artifacts (the closing edge $j = K_i - 1$ is included). The cross-product, area, and centroid are computed element-wise and reduced over the vertex dimension in a single kernel launch.

Containment Verification and Exterior Centroid Correction.

For convex polygons the area-weighted centroid always lies inside the boundary. For the mildly concave nuclei encountered in histopathology, it may fall outside. A crossing-number point-in-polygon test is applied: a horizontal ray is cast in the $+X$ direction, and the number of edge crossings determines interiority (odd crossings = inside).

When the centroid falls outside the polygon, a binary search positions it at an interior point. The target $\mathbf{T}$ is the mean of the three real vertices closest to the exterior centroid in Euclidean distance:

$$\mathbf{T} = \frac{1}{3} (\mathbf{V}_{j_1} + \mathbf{V}_{j_2} + \mathbf{V}_{j_3}) \quad (3-5)$$

The algorithm iteratively probes the line segment $[\mathbf{P}, \mathbf{T}]$ at fractions $\alpha \in \{0.5, 0.75, 0.875, 0.9375\}$ testing each candidate $\mathbf{Q}_\alpha = \mathbf{P} + \alpha(\mathbf{T} - \mathbf{P})$ for interiority via the crossing-number test. The candidate is updated only if the test succeeds. After 5 iterations the final precision is $1 - 0.96875 \approx 3.1\%$ of $\|\mathbf{T} - \mathbf{P}\|$. In practice, histopathology nuclei exhibit at most mild concavity, so the area-weighted centroid is interior in nearly all cases and the fallback is rarely triggered.

Ray–Edge Intersection.

The core computation solves for the intersection of each ray with every polygon edge. The ray from centroid $\mathbf{P}$ in unit direction $\mathbf{D}_r = (\cos \theta_r, \sin \theta_r)^\top$ is parameterized as:

$$\mathbf{L}_r(t) = \mathbf{P} + t \mathbf{D}_r, \quad t \geq 0 \quad (3-6)$$

An edge segment from $\mathbf{V}_j$ to $\mathbf{V}_{j+1}$ with edge vector $\mathbf{E}_j = \mathbf{V}_{j+1} - \mathbf{V}_j$ is:

$$\mathbf{Q}_j(s) = \mathbf{V}_j + s \mathbf{E}_j, \quad s \in [0, 1] \quad (3-7)$$

Setting $\mathbf{L}_r(t) = \mathbf{Q}_j(s)$ and defining the offset $\mathbf{F}_j = \mathbf{V}_j - \mathbf{P}$ yields a 2×2 linear system:

$$\underbrace{\begin{bmatrix} \cos \theta_r & -E_{j,x} \\ \sin \theta_r & -E_{j,y} \end{bmatrix}}_{\mathbf{M}_{rj}} \begin{bmatrix} t \\ s \end{bmatrix} = \underbrace{\begin{bmatrix} F_{j,x} \\ F_{j,y} \end{bmatrix}}_{\mathbf{b}_j} \quad (3-8)$$

The system is solved via Cramer's rule using the 2D cross product $\mathbf{a} \times \mathbf{b} = a_x b_y - a_y b_x$:

$$\det_{rj} = \det(\mathbf{M}_{rj}) = -\cos \theta_r \cdot E_{j,y} + \sin \theta_r \cdot E_{j,x} \quad (3-9)$$

$$t_{rj} = \frac{-\mathbf{F}_j \times \mathbf{E}_j}{\det_{rj}}, \quad s_{rj} = \frac{\mathbf{D}_r \times \mathbf{F}_j}{\det_{rj}} \quad (3-10)$$

The right-hand side $\mathbf{b}_j$ is independent of the ray direction; only the first column of $\mathbf{M}_{rj}$ varies with r . Each (r, j) pair is solved independently, enabling full data-parallelism over all three dimensions (polygon, ray, edge) simultaneously.

Validity Filtering and Nearest-Hit Selection.

An intersection is valid if and only if: (1) $|\det_{rj}| > \epsilon_d$ (ray is not parallel to the edge, $\epsilon_d = 10^{-10}$), (2) $t_{rj} > 0$ (intersection lies in front of the centroid; the strict inequality prevents self-intersection artifacts when the centroid lies exactly on the boundary), (3) $-\epsilon \leq s_{rj} \leq 1 + \epsilon$ (intersection falls on the edge segment, $\epsilon = 10^{-6}$), and (4) the edge is a real polygon edge (not a padding artifact). Degenerate cases are handled as follows:

- **Ray through shared vertex:** both adjacent edges produce valid hits with the same t ; the minimum selection naturally resolves this.

- **Ray tangent to edge:** filtered by the determinant threshold $|\det_{rj}| < \epsilon_d$.
- **Ray parallel but offset:** produces $t \rightarrow \pm\infty$, filtered by the $t > 0$ condition.
- **Ray collinear with edge:** the edge lies along the ray direction; d_r is conservatively set to $d_r = 0$.

For each ray r , the boundary distance is the minimum positive t among valid intersections:

$$d_r = \min_{j \in \mathcal{V}_r} t_{rj}, \quad \mathcal{V}_r = \{j : \text{valid}_{rj} = \text{True}\} \quad (3-11)$$

Since $\mathbf{D}_r$ is a unit vector, $d_r = t_{rj^*}$ is directly the Euclidean distance from the centroid to the boundary intersection. Rays with no valid intersection ($\mathcal{V}_r = \emptyset$) are assigned $d_r = 0$, following the convention that zero distance indicates a missed ray.

Batched GPU Implementation.

The entire pipeline is implemented as vectorized tensor operations in PyTorch, processing all N polygons in a region of interest simultaneously on GPU. Polygons are padded to $K_{\max}$ vertices by repeating the last vertex; past the real vertex count, the padding creates degenerate zero-length edges that naturally fail the determinant check and are excluded from results.

The core intersection solve operates over the full $(N, R, K_{\max})$ tensor. Input shapes are broadcast as follows: vertex coordinates $(N, 1, K_{\max})$, centroid $(N, 1, 1)$, ray trig functions $(1, R, 1)$, expanding element-wise to $(N, R, K_{\max})$. All Cramer's rule computations, validity masking, and the final min-reduction over the edge dimension execute in a single kernel launch without explicit loops.

The computational work scales as $O(N \cdot R \cdot K_{\max})$. The only serial component is the exterior centroid fallback (5 binary search iterations), which operates on a negligible fraction of the batch since histopathology nuclei rarely produce exterior centroids. For a typical region of interest ($N \approx 1000$, $R = 32$, $K_{\max} \approx 500$), the intersection tensors require approximately $4 \cdot N \cdot R \cdot K_{\max} \approx 64\text{MB}$ each, well within modern GPU memory.

Inverse Decoding.

The raycast representation decodes back to a polygon by reconstructing vertex positions from the centroid and radial distances:

$$\mathbf{V}_r = \mathbf{P} + d_r \cdot \mathbf{D}_r, \quad r = 0, \dots, R - 1 \quad (3-12)$$

connected cyclically. In batched form for N annotations with distance matrix $\mathbf{D} \in \mathbb{R}^{N \times R}$ and precomputed direction vectors $\mathbf{c} = (\cos \theta_0, \dots, \cos \theta_{R-1})$ and $\mathbf{s} = (\sin \theta_0, \dots, \sin \theta_{R-1})$:

$$\mathbf{V}_x = c_x[:, \text{None}] + \mathbf{D} \odot \mathbf{c}^\top, \quad \mathbf{V}_y = c_y[:, \text{None}] + \mathbf{D} \odot \mathbf{s}^\top \quad (3-13)$$

yielding vertex coordinates $\mathbf{V} \in \mathbb{R}^{N \times R \times 2}$. The angular constants are precomputed once and shared across ingestion, model inference, loss computation, and evaluation.

Data Ingestion Design.

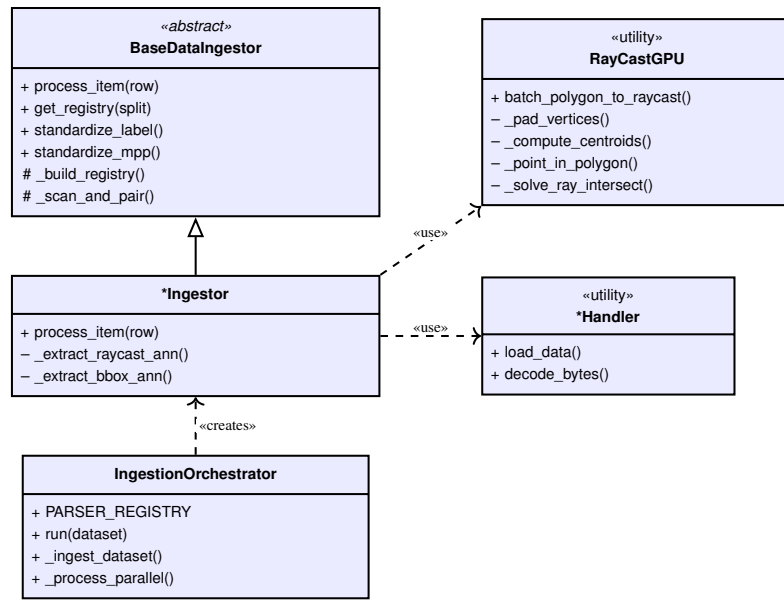


Figure 3.5. UML class diagram of the modular ingestion pipeline. `BaseDataIngestor` defines the shared infrastructure; concrete ingestors implement format-specific `process_item` logic; all converge on the unified raycast annotation format via `RayCastGPU`.

A practical computational pathology system must ingest annotations from heterogeneous source formats: binary instance masks, polygon coordinates, comma-separated coordinate strings, and integer label maps. Rather than implementing a monolithic converter per dataset, the ingestion pipeline follows a modular architecture with three design principles.

First, **format-agnostic core logic** (centroid computation, ray–edge intersection solving, MPP harmonization, and label translation) is shared across all ingestors. Second, each concrete ingestor implements only the **format-specific extraction** logic that parses its source format into the common intermediate representation: polygon vertex arrays with class labels. Third, **configuration-driven routing** specifies the choice of ingestor, split

strategy, modality pairing, and label mapping declaratively, without code changes.

The architecture is built around an abstract base class that defines the contract every ingestor must fulfill: for each region of interest, produce a unified output $(id_{\text{roi}}, \mathbf{I}, \mathbf{A}, t_{\text{tissue}})$ consisting of an identifier, an RGB image, a raycast annotation matrix $\mathbf{A} \in \mathbb{R}^{N \times (3+R)}$, and a tissue provenance label. The sole abstract method is `process_item`, which each concrete ingestor implements for its specific format.

The base class provides a template method for **registry construction** in three phases: (1) file discovery and pairing, where image files are matched with annotation files using either bundled-archive (same file) or physical-parallel (paired directories) strategies; (2) split assignment, where each pair is assigned to train/val/test using physical subdirectory names, filename regex patterns, or stratified random sampling by tissue type; and (3) registry assembly into a work queue. A **two-step label translation** decouples source vocabularies from the global integer encoding: a namespace map $\mathcal{N}$ translates raw labels to standardized strings shared across all datasets, and a global cell map $\mathcal{G}$ maps standardized strings to integer class IDs. Adding a new dataset requires only a new namespace map. Similarly, all annotations are rescaled to a common target microns-per-pixel via **MPP harmonization**, scaling image dimensions and annotation coordinates uniformly by the ratio $\rho = \sigma_{\text{native}} / \sigma_{\text{target}}$.

Concrete ingestors differ only in how they extract polygon vertices: mask-based sources decode compressed byte strings via Moore-neighbor contour tracing, while vector-based sources read polygon coordinates directly from geometry arrays or comma-separated coordinate strings requiring no rasterization. Mask-based ingestors process ROIs in parallel using a process pool, deferring GPU ray-cast conversion to the main process to avoid CUDA cross-process deadlock. All ingestors converge through a single conversion entry point, `RayCastGPU.batch_polygon_to_raycast`, which receives the format-independent intermediate representation (variable-length vertex arrays and class labels) and produces the unified raycast annotation $\mathbf{A}$.

The `IngestionOrchestrator` governs the end-to-end process: a string-keyed registry maps configuration keys to concrete ingestor classes, validated at both config-load time and runtime. A two-layer YAML hierarchy (global defaults overridden per dataset) enables declarative specification of all parameters. Row-level errors are caught and logged without terminating the remaining dataset, so a single corrupt ROI does not halt ingestion. This separation of concerns means that adding a new dataset typically requires only a YAML configuration entry with a namespace map and format key, while the computational core remains invariant across all sources.

3.2.1.2 Transform Pipeline

Data Profile Registration.

Before any per-tile transformation is applied, the ingestion pipeline registers a data profile for each image in the dataset. This profile captures three categories of metadata that downstream transforms consume: (1) **tissue type and dataset source**, recorded as a string identifier and integer encoding, enabling stratified sampling across datasets and tissue categories during population-level stain estimation; (2) **native image dimensions** (W, H) and **microns-per-pixel** (MPP), required by the size standardization step to compute chopping and padding parameters; and (3) **tile-level stain statistics**, including a boolean flag indicating whether the tile contains sufficient tissue content (at least 100 non-white pixels) to be eligible for stain vector estimation. Profiles are assembled during the ingestion registry phase and serialized alongside the raycast annotations, so that the transform pipeline can load them without re-reading the source images. This separation ensures that the transform pipeline operates purely on pre-computed metadata rather than performing redundant I/O.

Stain Vector Estimation and Normalization.

Histopathology images exhibit substantial color variation across scanners, staining batches, and laboratories. A model trained on darkly-stained slides may perform poorly on lightly-stained ones. The transform pipeline addresses this through a three-phase process. First, images are chopped into $S \times S$ tiles. Second, a subset of tiles is Pareto-sampled, the Macenko method [16] estimates per-tile stain vectors, and the results are aggregated into a population-level profile. Third, every tile is normalized against this profile and padded to a uniform size.

Population-Level Stain Vector Estimation.

Rather than processing every tile – which wastes computation and allows large datasets to dominate – stratified Pareto sampling is applied following Agraz et al. [32], who showed that 20% of a cohort suffices for a representative profile. The chunked registry is partitioned into strata by dataset and tissue type, ensuring each dataset contributes proportionally and rare tissue types are represented. Within each stratum containing n_j tiles, the number sampled is:

$$k_j = \max(1, \lceil 0.20 \cdot n_j \rceil) \quad (3-14)$$

Tiles are drawn uniformly at random with a fixed seed. The estimation set $\mathcal{S}$ is the union across all J strata.

For each sampled tile $k \in \mathcal{S}$, the Macenko method estimates a per-tile stain matrix $\mathbf{M}_{\text{stain}}^{(k)} \in \mathbb{R}^{2 \times 3}$ and maximum concentrations $\mathbf{c}_{\text{max}}^{(k)} \in \mathbb{R}^2$. The population profile is aggregated across the K tiles in $\mathcal{S}$ that yield valid estimates. The stain matrix is averaged since stain directions vary smoothly, while maximum concentrations use the 99th percentile to establish a robust upper bound:

$$\overline{\mathbf{M}} = \frac{1}{K} \sum_{k \in \mathcal{S}} \mathbf{M}_{\text{stain}}^{(k)}, \quad \mathbf{c}_{\text{max}}^* = P_{99}(\{\mathbf{c}_{\text{max}}^{(k)}\}_{k \in \mathcal{S}}) \quad (3-15)$$

Tiles that fail estimation (all-white or fewer than 100 tissue pixels) are silently excluded from K . The profile is serialized as JSON containing $\overline{\mathbf{M}}$, $\mathbf{c}_{\text{max}}^*$, the estimation method, and the tile count K for reuse during normalization and at inference time.

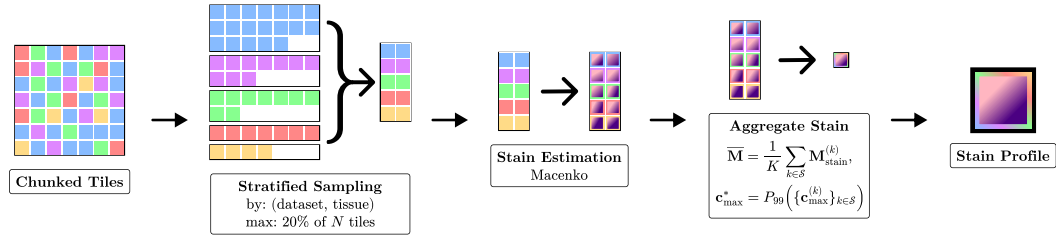


Figure 3.6. Population-level stain vector estimation. A stratified 20% subset of tiles is sampled per dataset and tissue type; the Macenko method estimates per-tile stain matrices and maximum concentrations, which are aggregated into the population profile $(\overline{\mathbf{M}}, \mathbf{c}_{\text{max}}^*)$.

Per-Tile Stain Normalization.

Given the population target $(\overline{\mathbf{M}}, \mathbf{c}_{\text{max}}^*)$ and a per-tile source estimate $(\mathbf{M}_{\text{source}}, \mathbf{c}_{\text{source}})$, normalization consists of five steps.

The tile is first converted from RGB to optical density space via the Beer–Lambert law:

$$\text{OD} = -\log_{10}\left(\frac{I_0 + 1}{I_o}\right) \quad (3-16)$$

where the $+1$ avoids $\log(0)$ at black pixels. The OD image is then decomposed into Hematoxylin and Eosin stain concentration maps using the pseudoinverse of the source stain matrix:

$$\mathbf{C}_{\text{source}} = \text{OD} \cdot \mathbf{M}_{\text{source}}^+ \quad (3-17)$$

Each channel’s concentration is rescaled to match the population target range by

the per-channel percentile ratio:

$$\mathbf{C}_{\text{norm}} = \mathbf{C}_{\text{source}} \odot \frac{\mathbf{c}_{\text{max}}^*}{\mathbf{c}_{\text{source}}} \quad (3-18)$$

A small epsilon (10^{-6}) prevents division by zero when a concentration channel is empty. The normalized OD is reconstructed using the population mean stain directions, replacing the source-specific color signature with the standardized one:

$$\text{OD}_{\text{norm}} = \mathbf{C}_{\text{norm}} \cdot \overline{\mathbf{M}} \quad (3-19)$$

Finally, the normalized OD is converted back to RGB and clipped to $[0, 255]$:

$$I_{\text{norm}} = I_o \cdot 10^{-\text{OD}_{\text{norm}}} - 1 \quad (3-20)$$

The full transform can be expressed compactly as:

$$I_{\text{norm}} = I_o \cdot 10^{-\left(\text{OD} \cdot \mathbf{M}_{\text{source}}^+ \odot \frac{\mathbf{c}_{\text{max}}^*}{\mathbf{c}_{\text{source}}}\right) \cdot \overline{\mathbf{M}}} - 1 \quad (3-21)$$

Image Size Standardization.

Every ingested image must be reduced to a fixed $S \times S$ canvas (defaulting to 256×256 pixels) suitable for batched GPU training. Two operations handle the two possible regimes: images larger than S in either dimension are **chopped** into overlapping tiles; images smaller than or equal to S are **padded** with white pixels at the bottom and right edges.

Chopping: Dynamic Overlap Distribution.

For an image dimension $L > S$ with minimum overlap fraction α_{overlap} , the number of tiles n is computed from the effective stride $\sigma = S - \lfloor S \cdot \alpha_{\text{overlap}} \rfloor$. Tile starting positions are distributed uniformly via $\text{linspace}(0, L - S, n)$, which distributes any excess overlap evenly across all seams rather than concentrating it at the boundary.

Annotation-Aware Raycast Splitting.

Each cell is assigned to exactly one tile via a centroid-owns-cell policy: a cell belongs to the tile containing its centroid (c_x, c_y) . The splitting procedure has four steps. First, centroids are filtered to retain only cells within the tile bounds. Second, centroid coordinates are shifted to crop-relative space: $c'_x = c_x - x_0$, $c'_y = c_y - y_0$. Third, for each

of the R rays, the distance to each of the four tile boundaries is computed. For a ray in direction $\mathbf{D}_r = (\cos \theta_r, \sin \theta_r)$, the boundary distances are:

$$d_{\text{right}} = \frac{w - c'_x}{\cos \theta_r}, \quad d_{\text{left}} = \frac{-c'_x}{\cos \theta_r}, \quad d_{\text{bottom}} = \frac{h - c'_y}{\sin \theta_r}, \quad d_{\text{top}} = \frac{-c'_y}{\sin \theta_r} \quad (3-22)$$

where terms with $|\cos \theta_r| \leq \varepsilon$ or $|\sin \theta_r| \leq \varepsilon$ are set to $+\infty$ (the ray is parallel to that boundary). The maximum permissible distance is $d_{\text{max}} = \min(d_{\text{right}}, d_{\text{left}}, d_{\text{bottom}}, d_{\text{top}})$, and each ray is clipped: $d'_r = \min(d_r, d_{\text{max}})$. Fourth, cells that lose too many rays to boundary clipping are discarded via a survival rate filter:

$$\frac{|\{r : d'_r > 0\}|}{R} \geq \tau_{\text{survival}} \quad (3-23)$$

where $\tau_{\text{survival}} = 0.5$. This prevents heavily truncated cells with unreliable shape representations from entering the training data. Figure 3.7 illustrates the centroid-owns-cell policy and ray clipping procedure.

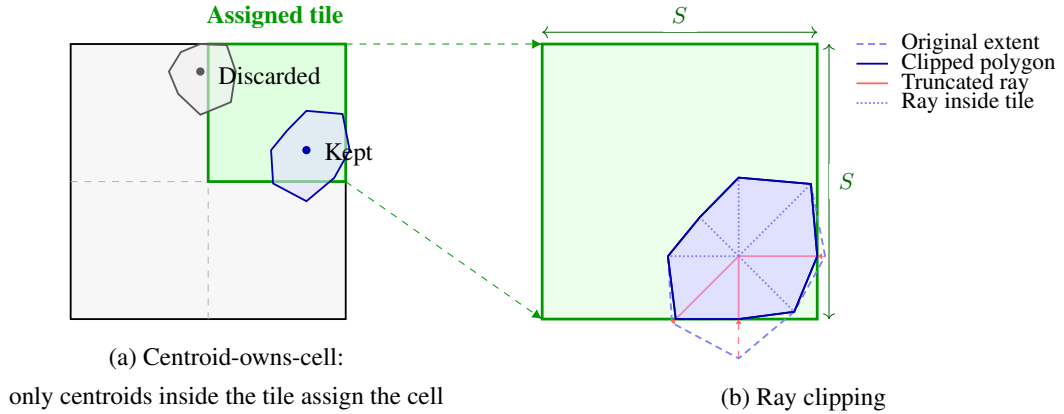


Figure 3.7. Annotation-aware raycast splitting. (a) The ROI is divided into overlapping tiles. Only cells whose centroids fall inside a given tile are assigned to it (centroid-owns-cell policy). (b) For each assigned cell, rays extending beyond the tile boundary are clipped to the nearest wall.

Padding.

After stain normalization, tiles with content smaller than $S \times S$ are padded with white pixels at the bottom and right edges. White padding is chosen because H&E slide backgrounds are white, so the model sees the same signal at tile edges as at slide borders. Bottom-right placement preserves annotation validity: the top-left origin $(0, 0)$ remains unchanged, so centroid positions and ray distances require no modification. The original content dimensions are recorded so that the training data loader can constrain random crop origins to the tissue-containing region.

3.2.1.3 Data Augmentation Strategies

RayCast Permutations.

Standard detection augmentations assume axis-aligned bounding boxes. Raycast annotations require specialized handling: spatial transforms (flips, rotations) permute the ray distance vector without changing magnitudes; scale multiplies all distances uniformly; and crops require ray clipping and a survival-rate filter to discard cells whose shape becomes unreliable.

All ray permutations depend only on R and the transform type, so they are pre-computed at configuration time:

Table 3.2. Precomputed ray permutation indices for each geometric transform.

Transform	Permutation	Example ($R = 32$)
Horizontal flip	$\pi_{\text{flip-h}}(r) = (R/2 - r) \bmod R$	ray 0 $\rightarrow$ 16
Vertical flip	$\pi_{\text{flip-v}}(r) = (R - r) \bmod R$	ray 0 $\rightarrow$ 0, ray 1 $\rightarrow$ 31
Rotate 90° CCW	$\pi_{\text{rot}}(r; 1) = (r + R/4) \bmod R$	ray 0 $\rightarrow$ 8
Rotate 180°	$\pi_{\text{rot}}(r; 2) = (r + R/2) \bmod R$	ray 0 $\rightarrow$ 16
Rotate 270° CCW	$\pi_{\text{rot}}(r; 3) = (r + 3R/4) \bmod R$	ray 0 $\rightarrow$ 24

Any augmentation that changes the spatial relationship between annotations and the canvas triggers a four-step clipping pipeline: (1) discard cells whose centroid falls outside the crop region, (2) translate surviving centroids to crop-relative coordinates, (3) for each ray, compute the distance to the four canvas boundaries and clip the ray to $d'_r = \min(d_r, d_{\text{max}}(r))$, and (4) discard cells with fewer than $\tau_{\text{surv}} = 0.3$ surviving rays. The boundary distance $d_{\text{max}}(r)$ is computed identically to the chopping procedure described above, using the same ray-by-ray intersection with tile edges.

Augmentation Operations.

The following augmentations are applied during training:

Random crop. Crop origin is constrained to the tissue-containing region to avoid sampling white padding. Annotations whose centroid falls outside the crop are dropped; remaining cells are translated to crop-relative space and rays are clipped.

Stain jitter. A photometric-only augmentation in HSV space that mimics inter-slide stain variation without modifying any annotation field:

$$H' = H + \delta_H, \quad S' = S \cdot \gamma_S, \quad V' = V \cdot \gamma_V \quad (3-24)$$

where $\delta_H \sim \mathcal{U}[-9^\circ, 9^\circ]$, $\gamma_S \sim \mathcal{U}[0.7, 1.3]$, and $\gamma_V \sim \mathcal{U}[0.8, 1.2]$. An optional Gaussian blur is applied with probability 0.2 ($\sigma \sim \mathcal{U}[0.5, 1.0]$).

Random scale. Isotropic scaling by $s \sim \mathcal{U}[0.7, 1.3]$: all spatial quantities (centroid, all ray distances) are multiplied by s . For $s > 1$, a random crop restores the image to $S \times S$; for $s \leq 1$, random padding places the scaled image on a white canvas. Boundary clipping follows.

Random translation. Shift by $(\Delta x, \Delta y)$ with $|\Delta x|, |\Delta y| \leq \lfloor 0.1 \cdot S \rfloor$. Only centroids move (ray distances are translation-invariant), followed by boundary clipping.

Horizontal flip. Applied with probability 0.5: centroid $c'_x = S - c_x$, and ray distances are permuted by the precomputed flip index.

3.2.2 Architecture Design

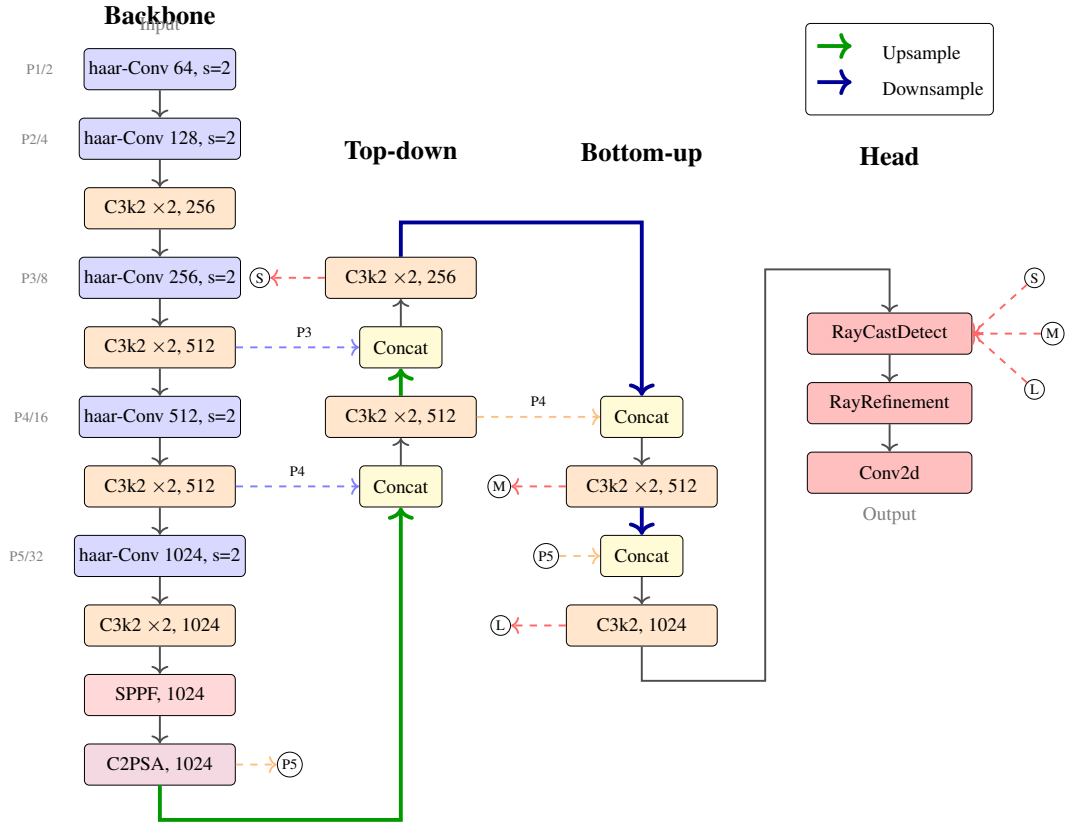


Figure 3.8. RayCastED model architecture

3.2.2.1 Proposed Block #1

3.2.2.2 Proposed Block #2

3.2.2.3 Proposed Block #3

3.2.3 Training Strategy and Optimization

3.2.3.1 Curriculum

explain about the dual assignment strategy where in the early epochs, the model mainly used the greedy TAL for one-to-many assignment, and as training progressed, it gradually shifted to use hungarian bipartite matching for one-to-one assignment. this curriculum-based strategy was adopted from YOLO26 with modification to the one-to-one branch. this training strategy was designed to let the model able to predict accurate detection without the need of post-processing steps such as NMS.

→ also explain the cost function used

3.2.3.2 Loss Functions

3.2.3.3 Auxiliary Heads

3.2.3.4 Optimization Algorithm and Hyperparameters

3.2.4 Evaluation Procedure

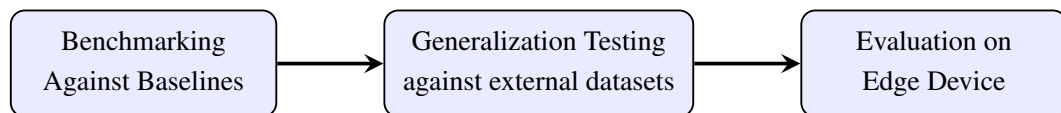


Figure 3.9. Evaluation Procedure Diagram

3.2.4.1 Evaluation Metrics

3.2.4.2 Benchmarking

3.2.4.3 Generalization Test

3.2.4.4 Evaluation on Edge Device

3.2.5 Inference Pipeline

TODO : add details

3.2.6 Deployment Architecture

3.2.6.1 Edge Device Specifications

3.2.6.2 Model Export Strategy

3.2.6.3 Environment Setup

3.3 Research Workflow

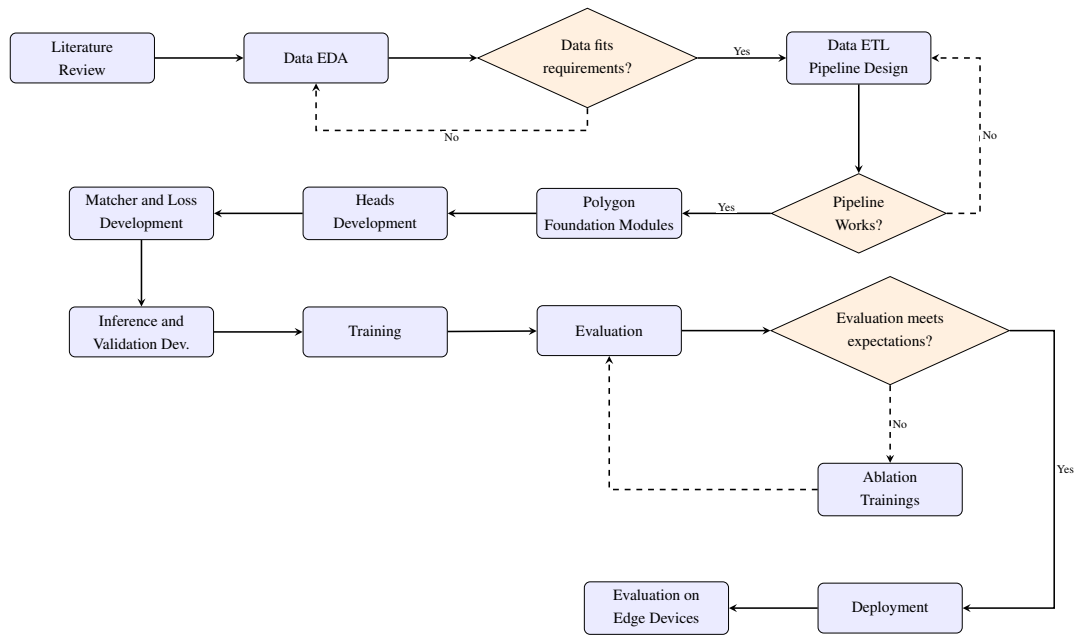


Figure 3.10. Research workflow diagram illustrating the iterative development process of the RayCastED model, from literature review and data preparation through model development, training, evaluation, and deployment.

Menguraikan prosedur yang akan digunakan dan jadwal atau alur penyelesaian setiap tahap. Alur penelitian ini dapat disajikan dalam bentuk diagram. Diagram dapat disusun dengan aturan yang baik semisal menggunakan *flowchart*. Aturan dan tutorial pembuatan *flowchart* dapat dilihat di <http://ugm.id/flowcharttutorial>. Setelah menggambarkannya, penulis wajib menjelaskan langkah-langkah setiap alur tugas akhir dalam sub bab tersendiri sesuai dengan kebutuhan.

CHAPTER IV

HASIL DAN PEMBAHASAN

Table 4.1. Evaluation Against LSP-DETR

Metric	LSP-DETR	RayCastED
F1	0.825	0.748
AJI	0.677	0.546
Precision	0.862	0.7618
Recall	0.791	0.7348
bPQ	0.657	0.5769
mPQ	0.476	0.4381
Params	45.0M	2.99M
Inference Time	-	3.75ms

Berikut ini adalah yang perlu diperhatikan untuk mengisi bab hasil dan pembahasan:

1. Setiap rumusan masalah boleh memiliki lebih dari 1 tujuan.
2. Setiap subbab harus spesifik menjawab setiap tujuan yang dituliskan.
3. Setiap rumusan masalah boleh dijawab dengan 1 subbab atau lebih.

Berikut ini adalah contoh sub bab untuk menjelaskan tujuan penelitian.

4.1 Pembahasan Tujuan 1 dengan Hasil Penelitian 1 (Ubah Judul Sesuai dengan Hal yang Hendak dibahas)

Sub bab pertama adalah membahas tujuan penelitian pertama dengan hasil penelitian ke-1. Dapat ditambahkan beberapa sub bab jika diperlukan.

4.2 Pembahasan Tujuan 1 dengan Hasil Penelitian 2 (Ubah Judul Sesuai dengan Hal yang Hendak dibahas)

Sub bab kedua adalah membahas tujuan penelitian pertama dengan hasil penelitian ke-2. Sub bab ini merupakan contoh tambahan sub bab pertama.

4.3 Pembahasan Tujuan 2 dengan Hasil Penelitian 3 (Ubah Judul Sesuai dengan Hal yang Hendak dibahas)

Sub bab ketiga adalah membahas tujuan penelitian kedua. Dapat ditambahkan beberapa sub bab jika diperlukan.

4.4 Perbandingan Hasil Penelitian dengan Hasil Terdahulu

Pembahasan penutup dapat menjelaskan mengenai kelebihan hasil pengembangan / penelitian dan kekurangan dibandingkan dengan skripsi atau penelitian terdahulu atau perbandingan terhadap produk lain yang ada di pasaran. Penulis dapat menggunakan tabel untuk membandingkan secara gamblang dan menjelaskannya.

CHAPTER V

TAMBAHAN (OPSIONAL)

Anda boleh menambahkan Bab jika diperlukan. Jumlah Bab tidak harus sesuai dengan *template*.

Bab tambahan ini diperlukan jika hasil penelitian untuk menjawab tujuan cukup panjang atau terdiri dari banyak sub bab. Mahasiswa boleh menjawab 1 tujuan penelitian dengan 1 bab.

CHAPTER VI

KESIMPULAN DAN SARAN

6.1 Kesimpulan

Kesimpulan dapat diawali dengan apa yang dilakukan dengan tugas akhir ini lalu dilanjutkan dengan poin-poin yang menjawab tujuan penelitian, apakah tujuan sudah tercapai atau belum, tentunya berdasarkan data ataupun hasil dari Bab pembahasan sebelumnya. Dalam beberapa hal, kesimpulan dapat juga berisi tentang temuan/*findings* yang Anda dapatkan setelah melakukan pengamatan dan atau analisis terhadap hasil penelitian.

Kesimpulan menjawab seberapa jauh rumusan masalah tercapai berdasarkan hasil penelitian. Semua rumusan masalah harus disimpulkan berdasarkan data penelitian.

6.2 Saran

Saran berisi hal-hal yang bisa dilanjutkan dari penelitian atau skripsi ini, yang belum dilakukan karena batasan permasalahan. Saran bukan berisi saran kepada sistem atau pengguna, tetapi saran diberikan kepada aspek penelitian yang dapat dikembangkan dan ditambahkan di penelitian atau skripsi selanjutnya.

Catatan: Mahasiswa perlu melihat sinkronisasi antara rumusan masalah, tujuan, metode, hasil penelitian, dan kesimpulan.

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Catatan: Daftar pustaka adalah apa yang dirujuk atau disitasi, bukan apa yang telah dibaca, jika tidak ada dalam sitasi maka tidak perlu dituliskan dalam daftar pustaka.

LAMPIRAN

L.1 Isi Lampiran

Lampiran bersifat opsional bergantung hasil kesepakatan dengan pembimbing dapat berupa:

1. Bukti pelaksanaan Kuesioner seperti pertanyaan kuesioner, resume jawaban responden, dan dokumentasi kuesioner.
2. Spesifikasi Aplikasi atau Sistem yang dikembangkan meliputi spesifikasi teknis aplikasi, tautan unduh aplikasi, manual penggunaan aplikasi, hingga screenshot aplikasi.
3. Cuplikan kode yang sekiranya penting dan ditambahkan.
4. Tabel yang terlalu panjang yang masih diperlukan tetapi tidak memungkinkan untuk ditayangkan di bagian utama skripsi.
5. Gambar-gambar pendukung yang tidak terlalu penting untuk ditampilkan di bagian utama. Akan tetapi, mendukung argumentasi/pengamatan/analisis.
6. Penurunan rumus-rumus atau pembuktian suatu teorema yang terlalu panjang dan terlalu teknis sehingga Anda berasumsi bahwa pembaca biasa tidak akan menelaah lebih lanjut. Hal ini digunakan untuk memberikan kesempatan bagi pembaca tingkat lanjut untuk melihat proses penurunan rumus-rumus ini.

LAMPIRAN

L.2 Panduan Latex

L.2.1 Syntax Dasar

L.2.1.1 Penggunaan Sitasi

Contoh penggunaan sitasi [33, 34] [35] [36] [37] [38, 39]

L.2.1.2 Penulisan Gambar

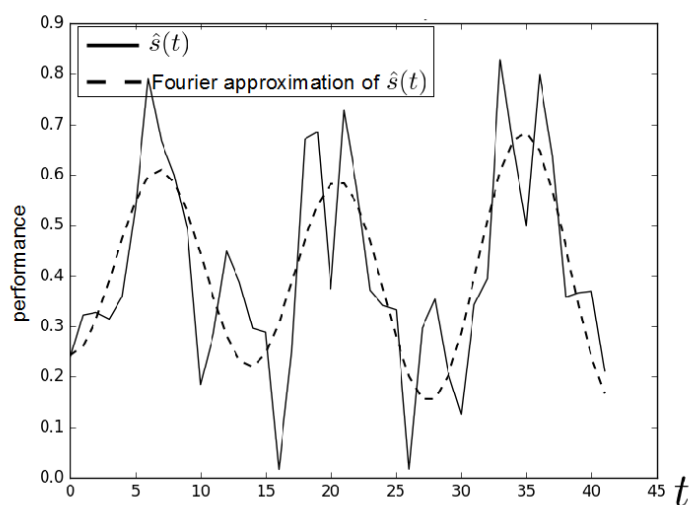


Figure L.1. Contoh gambar.

Contoh gambar terlihat pada Gambar L.1. Gambar diambil dari [39].

L.2.1.3 Penulisan Tabel

Table L.1. Tabel ini

ID	Tinggi Badan (cm)	Berat Badan (kg)
A23	173	62
A25	185	78
A10	162	70

Contoh penulisan tabel bisa dilihat pada Tabel L.1.

L.2.1.4 Penulisan formula

Contoh penulisan formula

$$L_{\psi_z} = \{t_i \mid v_z(t_i) \leq \psi_z\} \quad (1)$$

Contoh penulisan secara *inline*: $PV = nRT$. Untuk kasus-kasus tertentu, kita membutuhkan perintah "mathit" dalam penulisan formula untuk menghindari adanya jeda saat penulisan formula.

Contoh formula **tanpa** menggunakan "mathit": $PVA = RTD$

Contoh formula **dengan** menggunakan "mathit": $PVA = RTD$

L.2.1.5 Contoh list

Berikut contoh penggunaan list

1. First item
2. Second item
3. Third item

L.2.2 Blok Beda Halaman

L.2.2.1 Membuat algoritma terpisah

Untuk membuat algoritma terpisah seperti pada contoh berikut, kita dapat memanfaatkan perintah *algstore* dan *algrestore* yang terdapat pada paket *algcompatible*. Pada dasarnya, kita membuat dua blok algoritma dimana blok pertama kita simpan menggunakan *algstore* dan kemudian di-restore menggunakan *algrestore* pada algoritma kedua. Perintah tersebut dimaksudkan agar terdapat kesinamungan antara kedua blok yang sejatinya adalah satu blok.

Algorithm 1 Contoh algorima

```
1: procedure CREATESET( $v$ )  
2:   Create new set containing  $v$   
3: end procedure
```

Pada blok algoritma kedua, tidak perlu ditambahkan caption dan label, karena sudah menjadi satu bagian dalam blok pertama. Pembagian algoritma menjadi dua bagian ini berguna jika kita ingin menjelaskan bagian-bagian dari sebuah algoritma, maupun untuk memisah algoritma panjang dalam beberapa halaman.

```
4: procedure CONCATSET( $v$ )  
5:   Create new set containing  $v$   
6: end procedure
```

L.2.2.2 Membuat tabel terpisah

Untuk membuat tabel panjang yang melebihi satu halaman, kita dapat mengganti kombinasi *table* + *tabular* menjadi *longtable* dengan contoh sebagai berikut.

Table L.2. Contoh tabel panjang

header 1	header 2
foo	bar
foo	bar
foo	bar
foo	bar
foo	bar
foo	bar
foo	bar
foo	bar
foo	bar
foo	bar
foo	bar
foo	bar

L.2.2.3 Menulis formula terpisah halaman

Terkadang kita butuh untuk menuliskan rangkaian formula dalam jumlah besar sehingga melewati batas satu halaman. Solusi yang digunakan bisa saja dengan memindahkan satu blok formula tersebut pada halaman yang baru atau memisah rangkaian formula menjadi dua bagian untuk masing-masing halaman. Cara yang pertama mungkin akan menghasilkan alur yang berbeda karena ruang kosong pada halaman pertama akan diisi oleh teks selanjutnya. Sehingga di sini kita dapat memanfaatkan *align* yang sudah diatur dengan mode *allowdisplaybreaks*. Penggunaan *align* ini memungkinkan satu rangkaian formula terpisah berbeda halaman.

Contoh sederhana dapat digambarkan sebagai berikut.

$$\begin{aligned}
 x &= y^2 \\
 x &= y^3 \\
 a + b &= c \\
 x &= y - 2 \\
 a + b &= d + e \\
 x^2 + 3 &= y \\
 a(x) &= 2x
 \end{aligned}
 \tag{2}$$

$$b_i = 5x$$

$$10x^2 = 9x$$

$$2x^2 + 3x + 2 = 0$$

$$5x - 2 = 0$$

$$d = \log x$$

$$y = \sin x$$

LAMPIRAN

L.3 Format Penulisan Referensi

Penulisan referensi mengikuti aturan standar yang sudah ditentukan. Untuk internasionalisasi DTETI, maka penulisan referensi akan mengikuti standar yang ditetapkan oleh IEEE (*International Electronics and Electrical Engineers*). Aturan penulisan ini bisa diunduh di <http://www.ieee.org/documents/ieeecitationref.pdf>. Gunakan Mendeley sebagai *reference manager* dan *export* data ke format Bibtex untuk digunakan di Latex.

Berikut ini adalah sampel penulisan dalam format IEEE:

L.3.1 Book

Basic Format:

- [1] J. K. Author, "Title of chapter in the book," in Title of His Published Book, xth ed. City of Publisher, Country: Abbrev. of Publisher, year, ch. x, sec. x, pp. xxx-xxx.

Examples:

- [1] B. Klaus and P. Horn, Robot Vision. Cambridge, MA: MIT Press, 1986.
- [2] L. Stein, "Random patterns," in Computers and You, J. S. Brake, Ed. New York: Wiley, 1994, pp. 55-70.
- [3] R. L. Myer, "Parametric oscillators and nonlinear materials," in Nonlinear Optics, vol. 4, P. G. Harper and B. S. Wherret, Eds. San Francisco, CA: Academic, 1977, pp. 47-160.
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Polymers of Hexadromicon, J. Peters, Ed., 2nd ed. New York: McGraw-Hill, 1964, pp. 15-64.

L.3.2 Handbook

Basic Format:

[1] Name of Manual/Handbook, x ed., Abbrev. Name of Co., City of Co., Abbrev. State, year, pp. xx-xx.

Examples:

[1] Transmission Systems for Communications, 3rd ed., Western Electric Co., Winston Salem, NC, 1985, pp. 44-60.

[2] Motorola Semiconductor Data Manual, Motorola Semiconductor Products Inc., Phoenix, AZ, 1989.

[3] RCA Receiving Tube Manual, Radio Corp. of America, Electronic Components and Devices, Harrison, NJ, Tech. Ser. RC-23, 1992.

Conference/Prosiding

Basic Format:

[1] J. K. Author, "Title of paper," in Unabbreviated Name of Conf., City of Conf., Abbrev. State (if given), year, pp.xxx-xxx.

Examples:

[1] J. K. Author [two authors: J. K. Author and A. N. Writer] [three or more authors: J. K. Author et al.], "Title of Article," in [Title of Conf. Record as], [copyright year] © [IEEE or applicable copyright holder of the Conference Record]. doi: [DOI number]

Sumber Online/Internet

Basic Format:

[1] J. K. Author. (year, month day). Title (edition) [Type of medium]. Available: [http://www.\(URL\)](http://www.(URL))

Examples:

[1] J. Jones. (1991, May 10). Networks (2nd ed.) [Online]. Available: <http://www.atm.com>

Skripsi, Tesis dan Disertasi

Basic Format:

[1] J. K. Author, "Title of thesis," M.S. thesis, Abbrev. Dept., Abbrev. Univ., City of Univ., Abbrev. State, year.

[2] J. K. Author, "Title of dissertation," Ph.D. dissertation, Abbrev. Dept., Abbrev. Univ., City of Univ., Abbrev. State, year.

Examples:

[1] J. O. Williams, "Narrow-band analyzer," Ph.D. dissertation, Dept. Elect. Eng., Harvard Univ., Cambridge, MA, 1993. [2] N. Kawasaki, "Parametric study of thermal and chemical nonequilibrium nozzle flow," M.S. thesis, Dept. Electron. Eng., Osaka Univ., Osaka, Japan, 1993

LAMPIRAN

L.4 Contoh Source Code

L.4.1 Sample algorithm

Algorithm 2 Kruskal's Algorithm

```
1: procedure MAKESET( $v$ )
2:   Create new set containing  $v$ 
3: end procedure
4:
5: function FINDSET( $v$ )
6:   return a set containing  $v$ 
7: end function
8:
9: procedure UNION( $u, v$ )
10:  Unites the set that contain  $u$  and  $v$  into a new set
11: end procedure
12:
13: function KRUSKAL( $V, E, w$ )
14:   $A \leftarrow \{\}$ 
15:  for each vertex  $v$  in  $V$  do
16:    MakeSet( $v$ )
17:  end for
18:  Arrange  $E$  in increasing costs, ordered by  $w$ 
19:  for each  $(u, v)$  taken from the sorted list do
20:    if FindSet( $u$ )  $\neq$  FindSet( $v$ ) then
21:       $A \leftarrow A \cup \{(u, v)\}$ 
22:      Union( $u, v$ )
23:    end if
24:  end for
25:  return  $A$ 
26: end function
```

L.4.2 Sample Python code

```
1 import numpy as np
2
3 def incmatrix (genl1 , genl2):
4     m = len (genl1)
5     n = len (genl2)
6     M = None #to become the incidence matrix
7     VT = np.zeros ((n*m,1) , int) #dummy variable
8
9     #compute the bitwise xor matrix
10    M1 = bitxormatrix (genl1)
11    M2 = np.triu (bitxormatrix (genl2) ,1)
12
13    for i in range (m-1):
14        for j in range (i+1, m):
15            [r,c] = np.where (M2 == M1[i,j])
16            for k in range (len (r)):
17                VT[(i)*n + r[k]] = 1;
18                VT[(i)*n + c[k]] = 1;
19                VT[(j)*n + r[k]] = 1;
20                VT[(j)*n + c[k]] = 1;
21
22    if M is None:
23        M = np.copy (VT)
24    else:
25        M = np.concatenate ((M, VT) , 1)
26
27    VT = np.zeros ((n*m,1) , int)
28
29    return M
```


L.4.3 Sample Matlab code

```
1 function X = BitXorMatrix(A,B)
2 %function to compute the sum without charge of two vectors
3
4 %convert elements into unsigned integers
5 A = uint8(A);
6 B = uint8(B);
7
8 m1 = length(A);
9 m2 = length(B);
10 X = uint8(zeros(m1, m2));
11 for n1=1:m1
12     for n2=1:m2
13         X(n1, n2) = bitxor(A(n1), B(n2));
14     end
15 end
```